

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7698
Nombre de lignes df clean	7698
Nombre de features	59
% de NaN	0.0
Taille dataset entraînement	5388
Taille dataset test	2309
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_NH3_wl_12h_A_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_tem-

perature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_umidity_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D3_energy_rel, r_umidity_wl_12h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

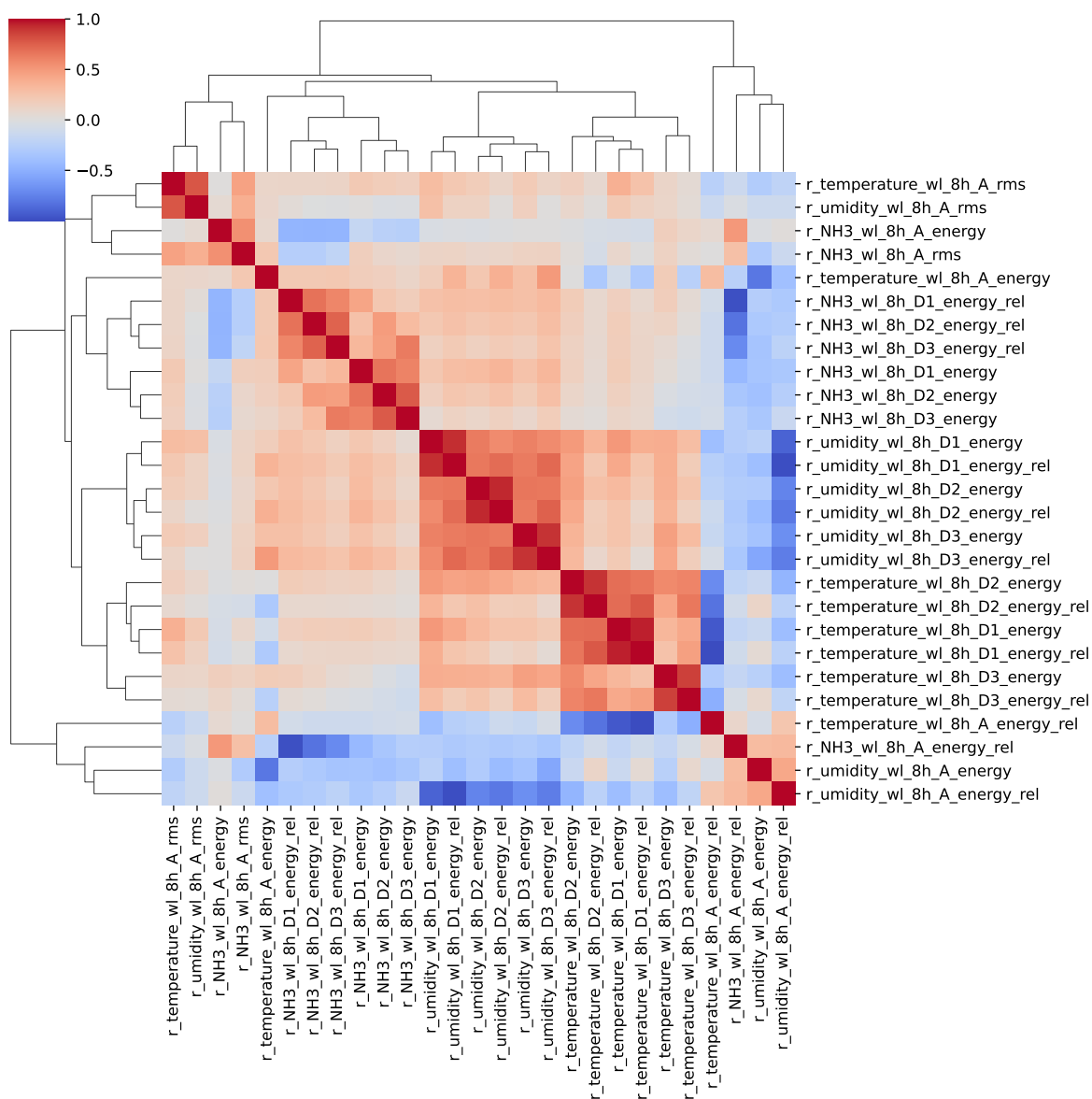
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_wl_8h_D1_energy ; **cluster 7:** r_NH3_wl_8h_D2_energy ; **cluster 8:** r_NH3_wl_8h_D3_energy ; **cluster 9:** r_NH3_wl_8h_A_energy ; **cluster 10:** r_NH3_wl_8h_A_rms ; **cluster 11:** r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D2_energy_rel ; **cluster 12:** r_NH3_wl_8h_D3_energy_rel ; **cluster 13:** r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D2_energy_rel ; **cluster 14:** r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_D3_energy_rel ; **cluster 15:** r_umidity_wl_8h_A_energy, r_temperature_wl_8h_A_energy ; **cluster 16:** r_temperature_wl_8h_A_rms ; **cluster 17:** r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D1_energy ; **cluster 18:** r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D2_energy ; **cluster 19:** r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_D3_energy ; **cluster 20:** r_umidity_wl_8h_A_rms ; **cluster 21:** r_NH3_wl_12h_D1_energy ; **cluster 22:** r_NH3_wl_12h_D2_energy ; **cluster 23:** r_NH3_wl_12h_D3_energy ; **cluster 24:** r_NH3_wl_12h_A_energy ; **cluster 25:** r_NH3_wl_12h_A_rms ; **cluster 26:** r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D1_energy_rel,

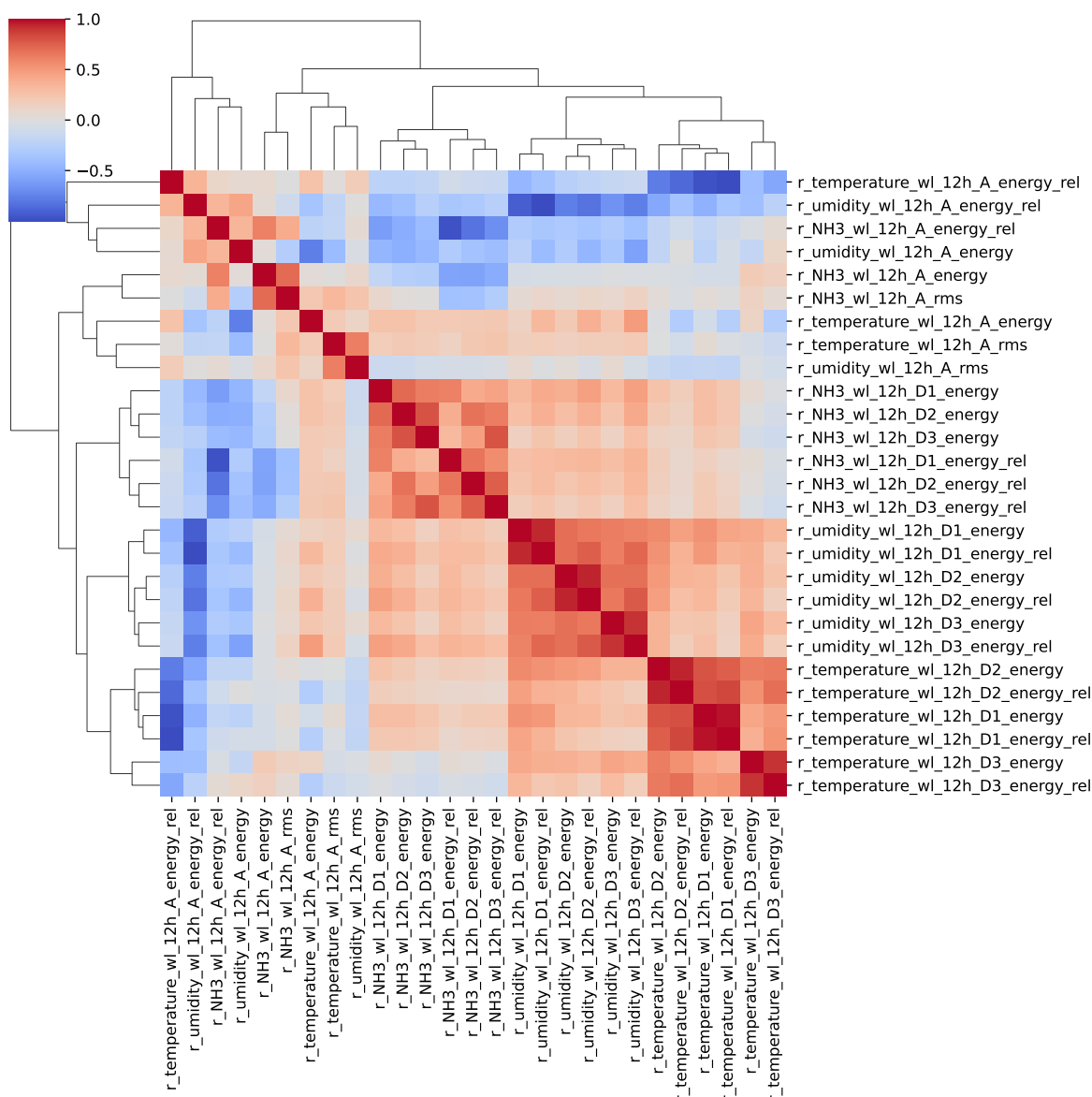
r_NH3_wl_12h_D2_energy_rel ; **cluster 27:** r_NH3_wl_12h_D3_energy_rel ; **cluster 28:** r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_D1_energy ; **cluster 29:** r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_D3_energy_rel ; **cluster 30:** r_temperature_wl_12h_A_energy ; **cluster 31:** r_temperature_wl_12h_A_rms ; **cluster 32:** r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_A_energy_rel ; **cluster 33:** r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_D3_energy_rel ; **cluster 34:** r_umidity_wl_12h_A_energy ; **cluster 35:** r_umidity_wl_12h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms





3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D3_energy_rel,

r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms

Len(Features_X): 35

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

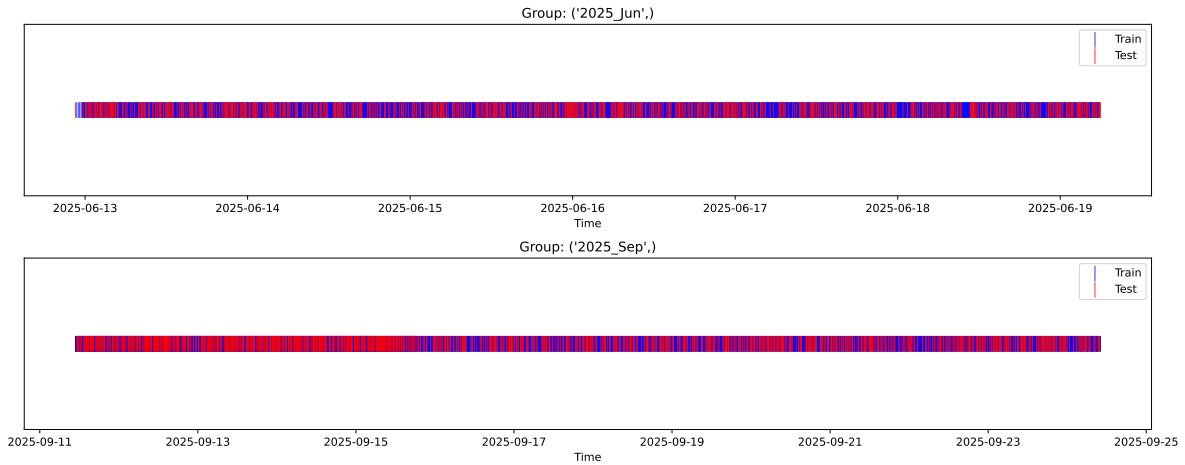
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

External Split: Train: 5388 samples Test: 2310 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



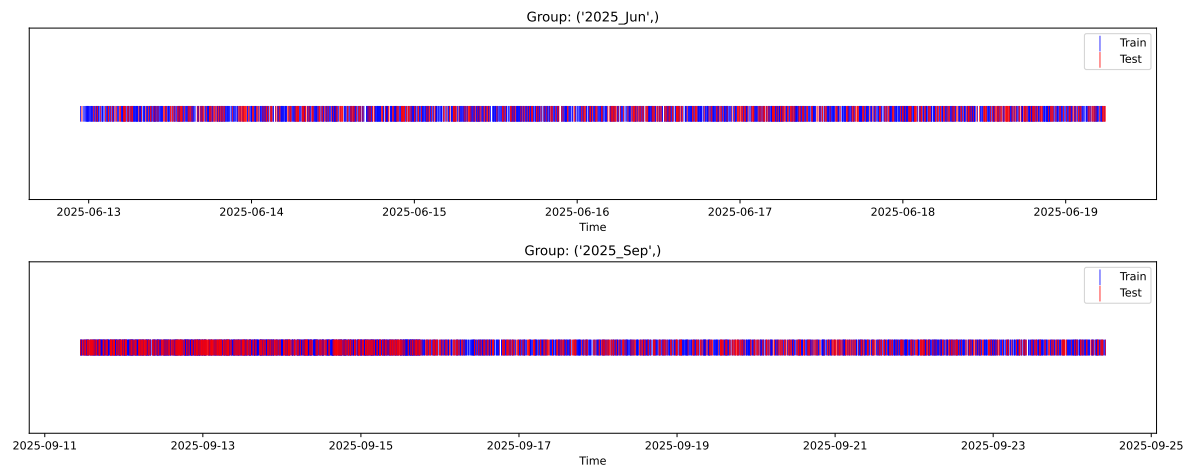
Fold 2/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



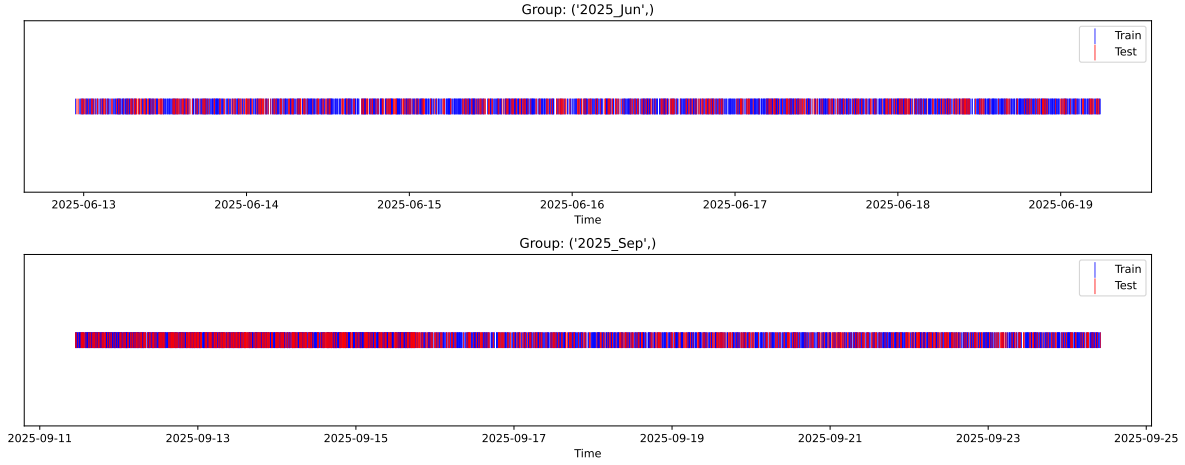
Fold 3/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



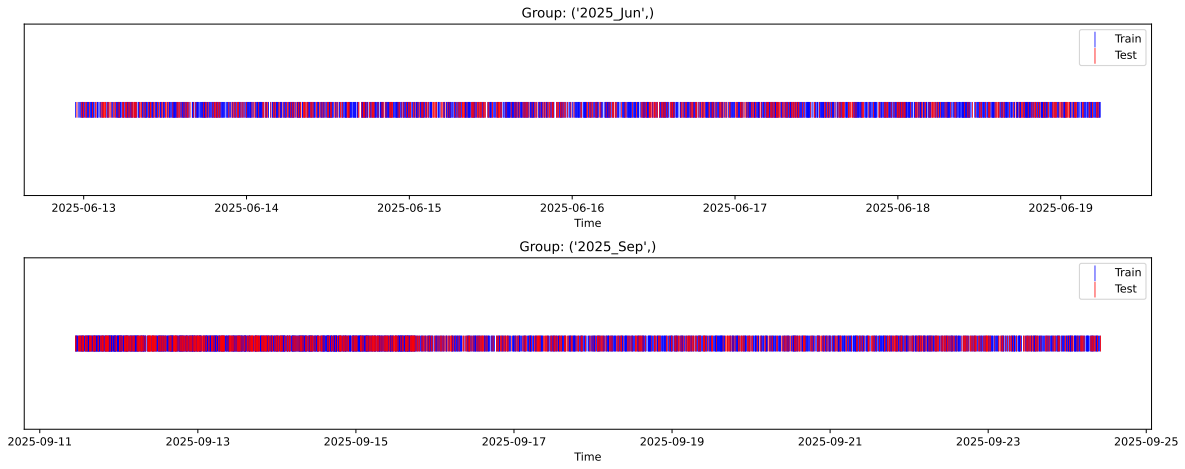
Fold 4/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.347, std=0.812, min=0.959, max=13.845
 y_{test} : mean=2.347, std=0.772, min=1.011, max=7.779

Fold 0 — y_{test} : mean=2.39, std=0.85, y_{train} : mean=2.33, std=0.79

Fold 1 — y_{test} : mean=2.33, std=0.80, y_{train} : mean=2.36, std=0.82

Fold 2 — y_{test} : mean=2.32, std=0.80, y_{train} : mean=2.36, std=0.81

Fold 3 — y_test: mean=2.36, std=0.85, y_train: mean=2.34, std=0.80

Fold 4 — y_test: mean=2.35, std=0.80, y_train: mean=2.34, std=0.82

train scores CV: [0.56047869 0.55527221 0.56186749 0.56865196 0.55344632]

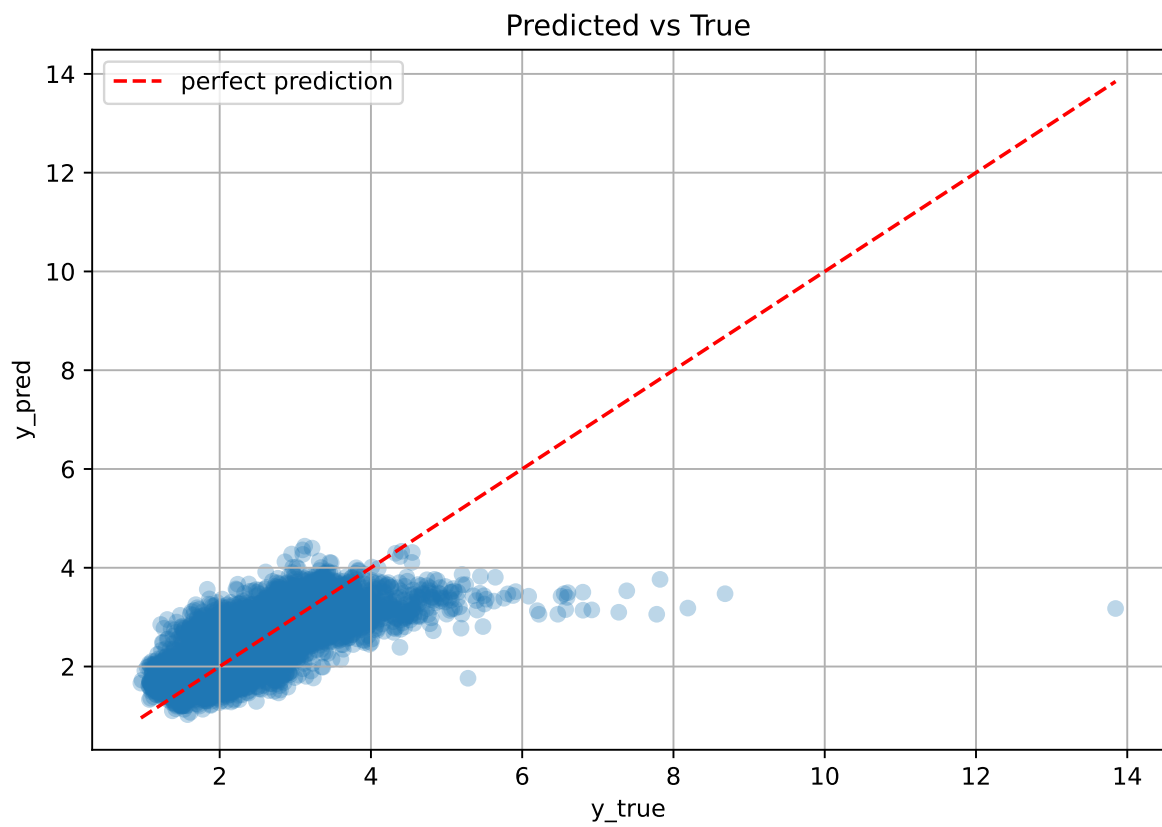
test scores CV: [0.54660531 0.56332069 0.5468406 0.53269506 0.56538484]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5599	0.551	0.0121	0.5499	0.5498	-0.0012	0.9979	1.0163

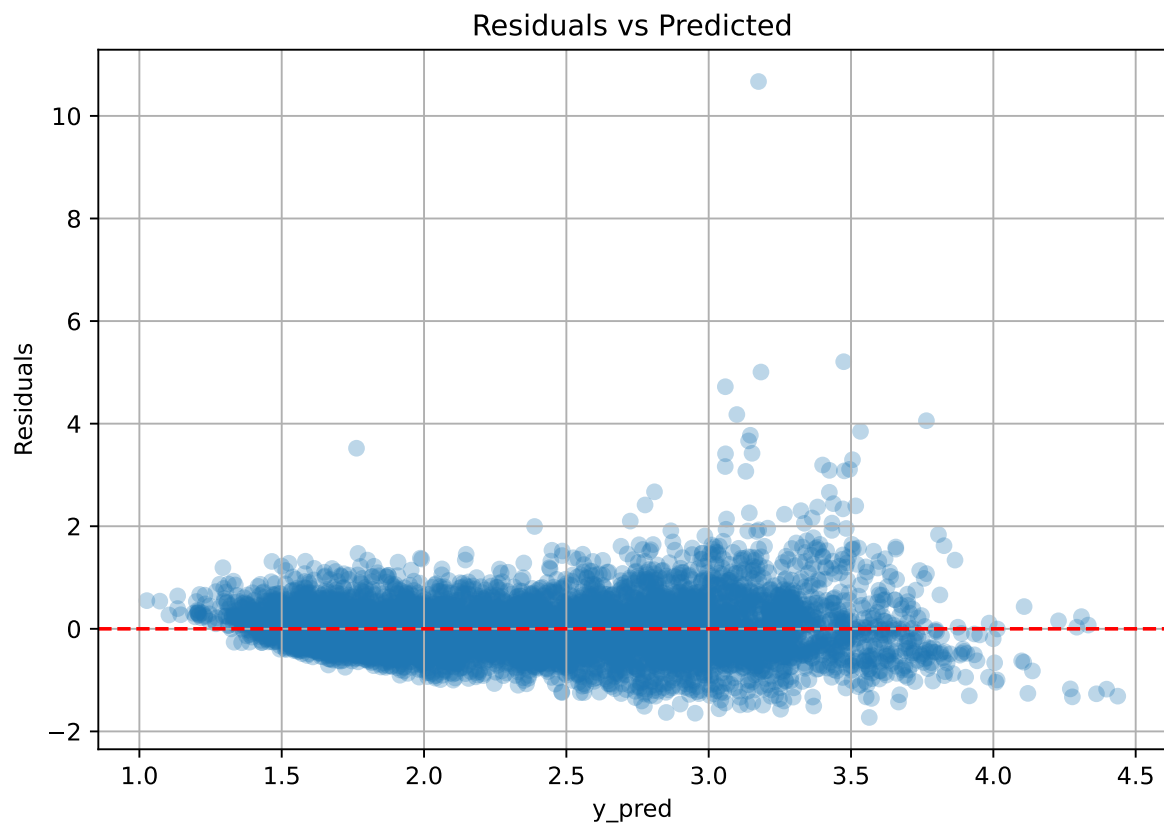
Coefficients:

coef_r_CO2 = -0.0534, coef_r_NH3 = 0.1372, coef_r_NH3_wl_12h_A_energy = -0.0724, coef_r_NH3_wl_12h_A_energy_rel = -0.0680, coef_r_NH3_wl_12h_A_rms = 0.0010, coef_r_NH3_wl_12h_D1_energy = -0.0101, coef_r_NH3_wl_12h_D2_energy = 0.0225, coef_r_NH3_wl_12h_D3_energy = -0.0279, coef_r_NH3_wl_12h_D3_energy_rel = -0.0129, coef_r_NH3_wl_8h_A_energy = -0.0867, coef_r_NH3_wl_8h_A_energy_rel = 0.0635, coef_r_NH3_wl_8h_A_rms = -0.0089, coef_r_NH3_wl_8h_D1_energy = 0.0451, coef_r_NH3_wl_8h_D2_energy = 0.0505, coef_r_NH3_wl_8h_D3_energy = -0.0339, coef_r_NH3_wl_8h_D3_energy_rel = -0.0332, coef_r_THI = 1.0058, coef_r_temperature = -1.0818, coef_r_temperature_wl_12h_A_energy = -0.0251, coef_r_temperature_wl_12h_A_energy_rel = -0.0154, coef_r_temperature_wl_12h_A_rms = 0.0123, coef_r_temperature_wl_12h_D3_energy = -0.0403, coef_r_temperature_wl_8h_A_energy_rel = -0.1520, coef_r_temperature_wl_8h_A_rms = -0.1220, coef_r_temperature_wl_8h_D3_energy = -0.0756, coef_r_umidity = -0.8475, coef_r_umidity_wl_12h_A_energy = -0.2603, coef_r_umidity_wl_12h_A_rms = -0.0862, coef_r_umidity_wl_12h_D2_energy_rel = 0.0064, coef_r_umidity_wl_12h_D3_energy = 0.0550, coef_r_umidity_wl_8h_A_energy = 0.4644, coef_r_umidity_wl_8h_A_energy_rel = 0.1001, coef_r_umidity_wl_8h_A_rms = 0.0226, coef_r_umidity_wl_8h_D2_energy_rel = -0.0179, coef_r_umidity_wl_8h_D3_energy_rel = 0.0890, intercept = 2.3473,

3.4.2 Scatter



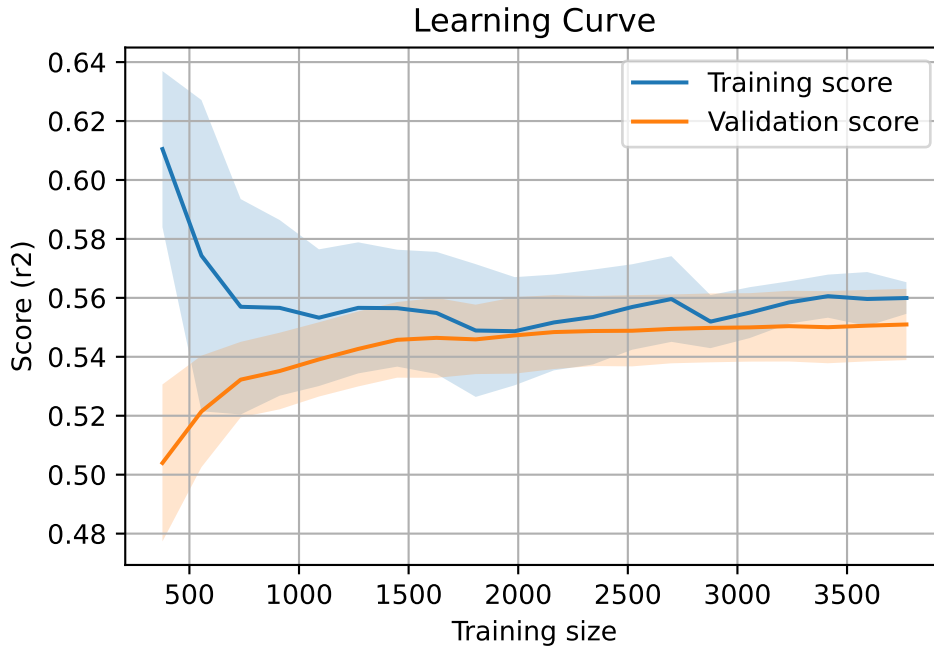
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7698, `Len(training)` : 5389,

`Len(training_real)` : 3773, `Len(test_of_training)` : 1616, `Len(test)` : 2309

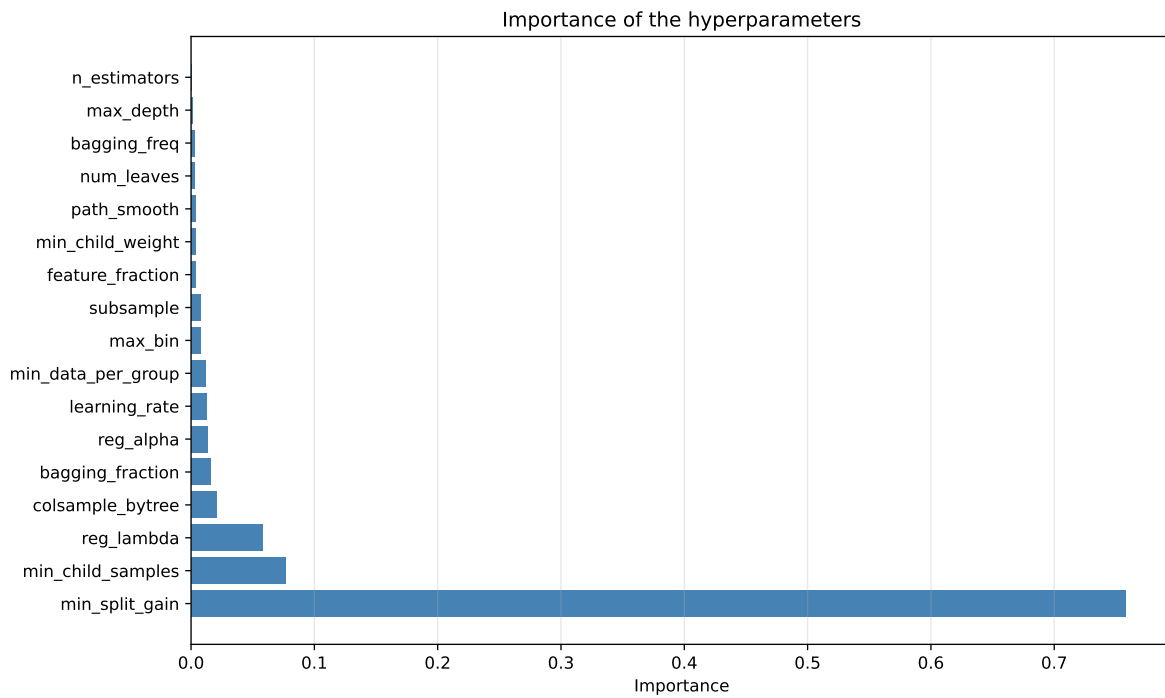
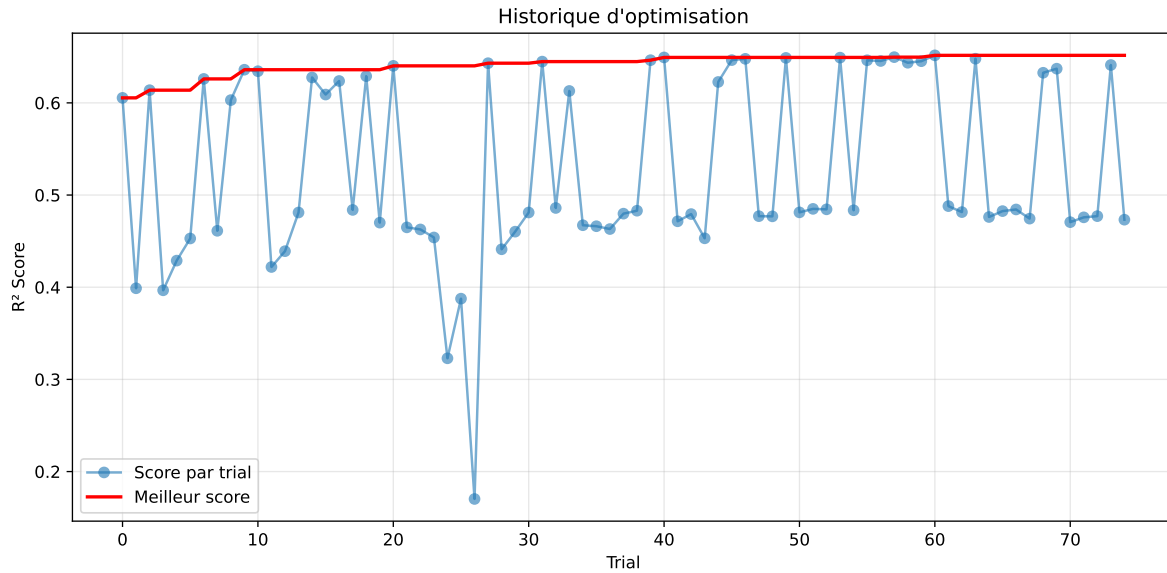


3.5 Model : LGBM

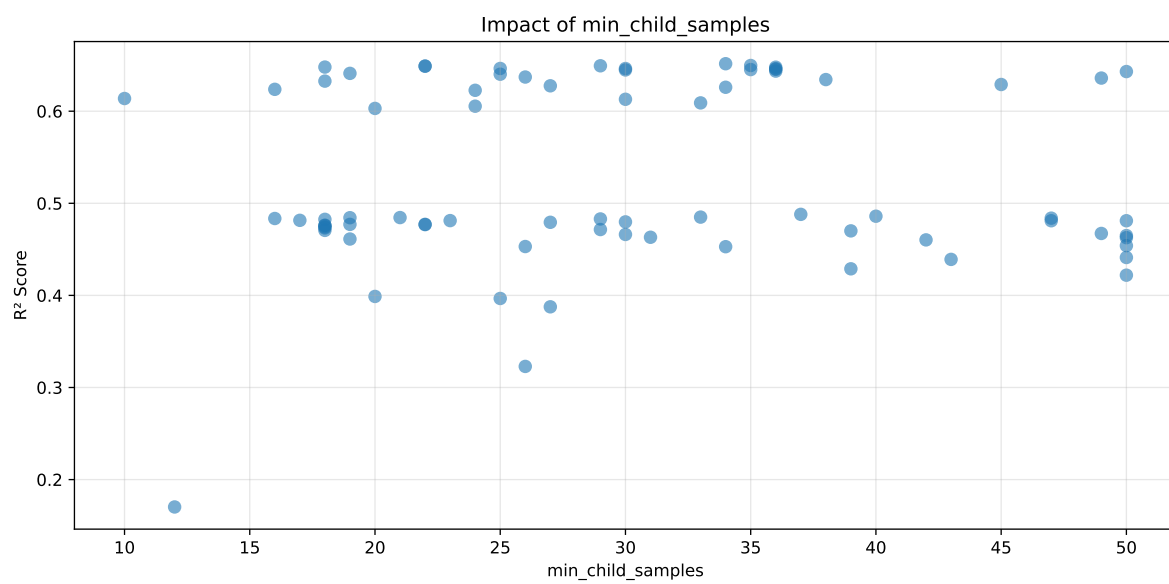
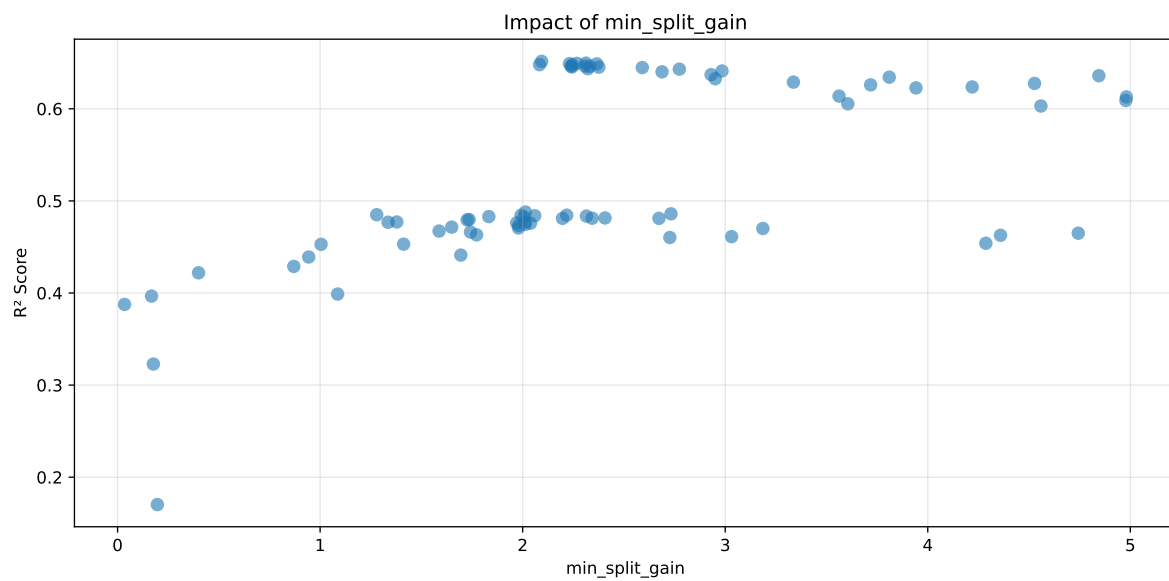
3.5.1 Gridsearch

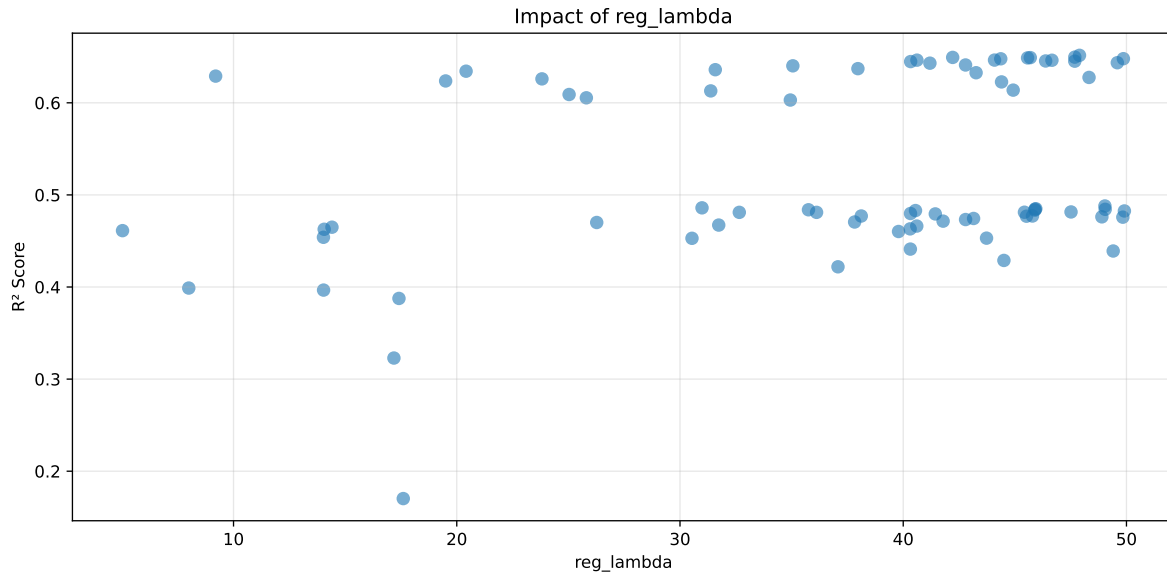
Distribution y_{train} vs y_{test} : y_{train} : mean=2.347, std=0.812, min=0.959, max=13.845
 y_{test} : mean=2.347, std=0.772, min=1.011, max=7.779

===== TOP Importance FEATURES =====
 feature importance r_{CO2} 105 $r_{umidity}$ 101 $r_{temperature_wl_12h_A_rms}$ 85 $r_{temperature_wl_12h_A_energy}$ 80 $r_{umidity_wl_12h_A_rms}$ 49 $r_{temperature}$ 39
 $r_{NH3_wl_8h_D1_energy}$ 36 r_{NH3} 35 $r_{NH3_wl_8h_A_energy}$ 34 $r_{temperature_wl_8h_A_energy_rel}$ 34 $r_{NH3_wl_8h_D3_energy_rel}$ 31 $r_{NH3_wl_12h_D1_energy}$ 30
 $r_{umidity_wl_12h_D3_energy}$ 30 r_{THI} 29 $r_{NH3_wl_12h_A_energy}$ 28 $r_{NH3_wl_12h_D3_energy}$ 23 $r_{NH3_wl_8h_D3_energy}$ 18 $r_{umidity_wl_8h_A_energy_rel}$ 18
 $r_{umidity_wl_8h_A_energy}$ 17 $r_{temperature_wl_8h_D3_energy}$ 17 $r_{NH3_wl_8h_D2_energy}$ 15 $r_{umidity_wl_12h_A_energy}$ 14 $r_{temperature_wl_12h_A_energy_rel}$ 13
 $r_{NH3_wl_12h_A_energy_rel}$ 12 $r_{temperature_wl_12h_D3_energy}$ 12 $r_{umidity_wl_8h_D2_energy_rel}$ 10 $r_{NH3_wl_12h_D3_energy_rel}$ 9
 $r_{NH3_wl_12h_A_rms}$ 9 $r_{NH3_wl_12h_D2_energy}$ 8 $r_{umidity_wl_12h_D2_energy_rel}$ 5 $r_{NH3_wl_8h_A_energy_rel}$ 5 $r_{NH3_wl_8h_A_rms}$ 3 $r_{umidity_wl_8h_D3_energy_rel}$ 3
 $r_{umidity_wl_8h_A_rms}$ 2 $r_{temperature_wl_8h_A_rms}$ 2



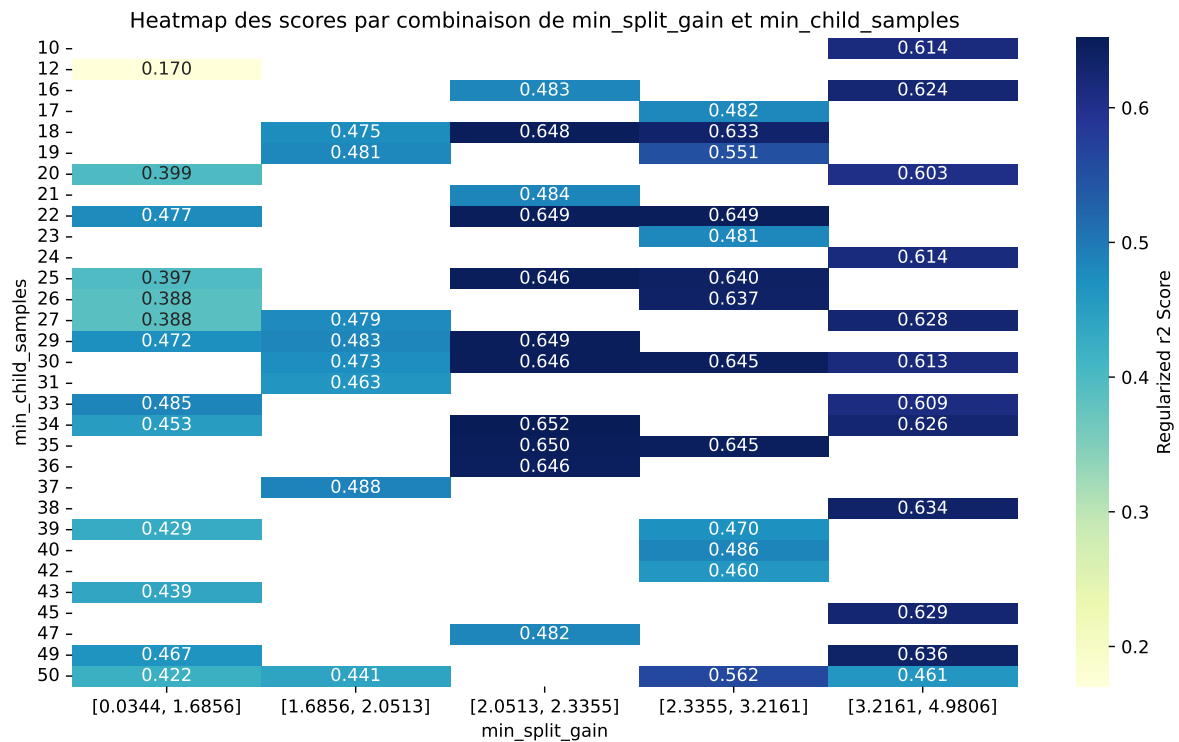
Paramètres avec >5% d'importance : ['min_split_gain', 'min_child_samples', 'reg_lambda']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.683	0.6516	0.6567	0.0214	0.6516	0.6819	0.0304	1.0466	1.0483
2	0.6806	0.6496	0.6543	0.0187	0.6496	0.6736	0.024	1.037	1.0476
3	0.6807	0.6493	0.6558	0.0218	0.6493	0.6773	0.028	1.0432	1.0484
4	0.6811	0.6491	0.6548	0.0194	0.6491	0.6784	0.0294	1.0452	1.0494
5	0.6795	0.6488	0.6561	0.0203	0.6488	0.6801	0.0313	1.0482	1.0473
6	0.6802	0.6479	0.6534	0.018	0.6479	0.6801	0.0322	1.0497	1.0498
7	0.6795	0.6478	0.6542	0.0186	0.6478	0.6718	0.024	1.037	1.049
8	0.6785	0.6464	0.6549	0.0213	0.6464	0.6749	0.0285	1.0442	1.0497
9	0.6778	0.6463	0.6536	0.0234	0.6463	0.6791	0.0328	1.0507	1.0487
10	0.6775	0.6462	0.6526	0.0241	0.6462	0.6761	0.0299	1.0462	1.0483
---	---	---	---	---	---	---	---	---	---
75	0.8832	0.7644	0.7788	0.0269	0.1703	0.7946	0.0302	1.0395	1.1554

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.04760552757516149, 'reg__num_leaves': 31, 'reg__subsample': 0.8715359677969046, 'reg__colsample_bytree': 0.7409569153919651, 'reg__min_child_samples': 34, 'reg__reg_alpha': 3.6787592028448275, 'reg__reg_lambda': 47.89583036791829, 'reg__min_split_gain': 2.0938567270767265, 'reg__path_smooth': 2.636291259738792, 'reg__min_child_weight': 6.186323570520597, 'reg__feature_fraction': 0.8475586311879236, 'reg__bagging_fraction': 0.6062115603428729, 'reg__bagging_freq': 5, 'reg__max_bin': 78, 'reg__min_data_per_group': 175}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.05119147087697007, 'reg__num_leaves': 31, 'reg__subsample': 0.8675977255112122, 'reg__colsample_bytree': 0.6185408290325224, 'reg__min_child_samples': 35, 'reg__reg_alpha': 3.3078333622762437, 'reg__reg_lambda': 47.6861900813999, 'reg__min_split_gain': 2.312291302961414, 'reg__path_smooth': 3.811248767305928, 'reg__min_child_weight': 2.110252950850309, 'reg__feature_fraction': 0.8499715025119103, 'reg__bagging_fraction': 0.6157508795861286, 'reg__bagging_freq': 5, 'reg__max_bin': 76, 'reg__min_data_per_group': 153}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.03865068731389409, 'reg__num_leaves': 15, 'reg__subsample': 0.8580085605861774, 'reg__colsample_bytree': 0.7938397286922205, 'reg__min_child_samples': 29, 'reg__reg_alpha': 2.1992490200213717, 'reg__reg_lambda': 42.21501034420541, 'reg__min_split_gain': 2.2677583338513294, 'reg__path_smooth': 1.3074547991612946, 'reg__min_child_weight': 9.548516930219956, 'reg__feature_fraction': 0.774312090867739, 'reg__bagging_fraction': 0.5518820824774577, 'reg__bagging_freq': 6, 'reg__max_bin': 89, 'reg__min_data_per_group': 189}

Rank 4: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.048604402014198286, 'reg__num_leaves': 31, 'reg__subsample': 0.8989997186458615, 'reg__colsample_bytree': 0.8123705558088286, 'reg__min_child_samples': 22, 'reg__reg_alpha': 3.223013846708753, 'reg__reg_lambda': 45.69201046170831, 'reg__min_split_gain': 2.2312248103902843, 'reg__path_smooth': 4.147547923981946, 'reg__min_child_weight': 2.1664240569990736, 'reg__feature_fraction': 0.8603509664876959, 'reg__bagging_fraction': 0.5959986969913917, 'reg__bagging_freq': 5, 'reg__max_bin': 63, 'reg__min_data_per_group': 171}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.04806659082440272, 'reg__num_leaves': 31, 'reg__subsample': 0.8942555436267734, 'reg__colsample_bytree': 0.8165018114843652, 'reg__min_child_samples': 22, 'reg__reg_alpha': 2.839386969899237, 'reg__reg_lambda': 45.570280285558795, 'reg__min_split_gain': 2.3659426948207187, 'reg__path_smooth': 4.314071273477197, 'reg__min_child_weight': 2.1661799936100765, 'reg__feature_fraction': 0.7871069013887777, 'reg__bagging_fraction': 0.5925225429286061, 'reg__bagging_freq': 5, 'reg__max_bin': 63, 'reg__min_data_per_group': 169}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.06514041373433838, 'reg__num_leaves': 31, 'reg__subsample': 0.8726129458262452, 'reg__colsample_bytree': 0.7401190054049834, 'reg__min_child_samples': 18, 'reg__reg_alpha': 4.658626072101967, 'reg__reg_lambda': 49.86266710734261, 'reg__min_split_gain': 2.083511899755242, 'reg__path_smooth': 4.958511199853722, 'reg__min_child_weight': 5.894529967177415, 'reg__feature_fraction': 0.787353195355772, 'reg__bagging_fraction': 0.6146820884421227, 'reg__bagging_freq': 4, 'reg__max_bin': 101, 'reg__min_data_per_group': 154}

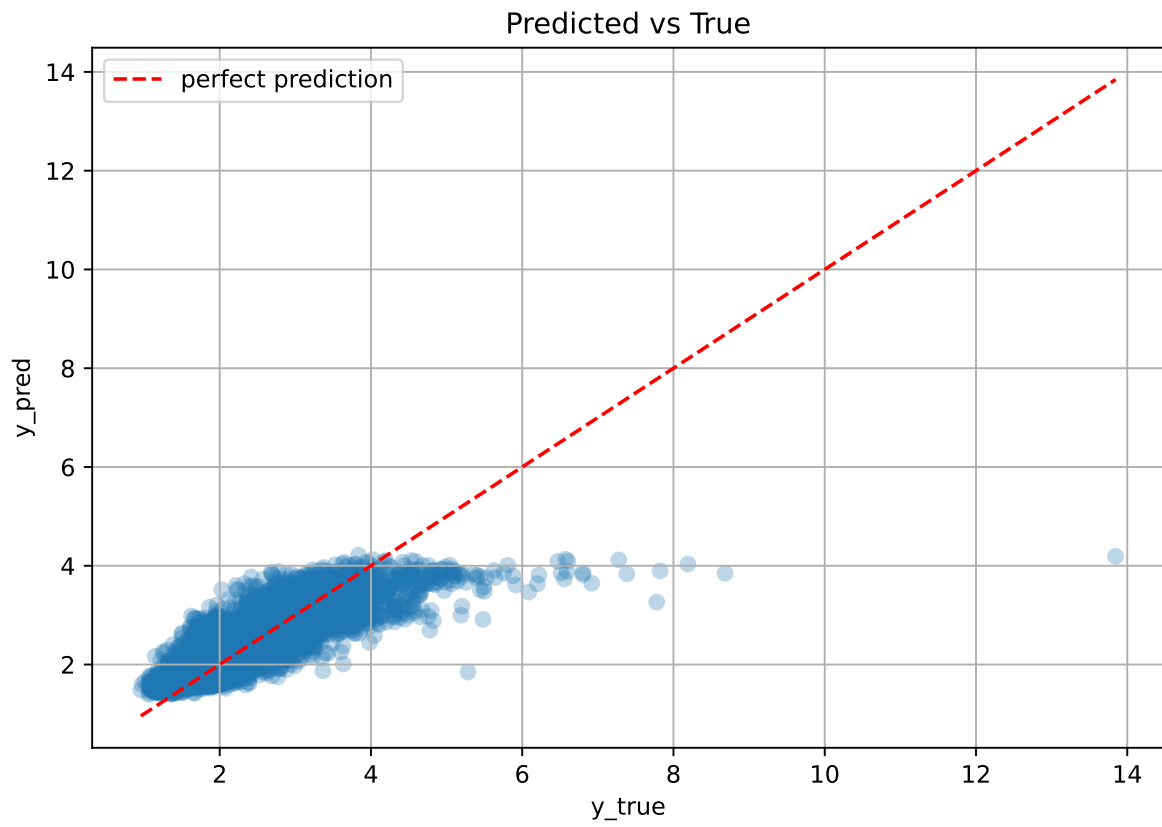
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.046444446038150386, 'reg__num_leaves': 15, 'reg__subsample': 0.8598232695389986, 'reg__colsample_bytree': 0.7666517793996787, 'reg__min_child_samples': 36, 'reg__reg_alpha': 2.8959837585033883, 'reg__reg_lambda': 44.36883747082108, 'reg__min_split_gain': 2.24579000359437, 'reg__path_smooth': 3.8917072514095263, 'reg__min_child_weight': 3.144024633221209, 'reg__feature_fraction': 0.7849600442109818, 'reg__bagging_fraction': 0.5822980447800764, 'reg__bagging_freq': 6, 'reg__max_bin': 69, 'reg__min_data_per_group': 169}

Rank 8: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.04779818158116617, 'reg__num_leaves': 31, 'reg__subsample': 0.8628589607131343, 'reg__colsample_bytree': 0.8088597431212612, 'reg__min_child_samples': 25, 'reg__reg_alpha': 2.3202413703694384, 'reg__reg_lambda': 44.089735664846884, 'reg__min_split_gain': 2.308118577285004, 'reg__path_smooth': 4.1415402404313095, 'reg__min_child_weight': 3.5860347735975595, 'reg__feature_fraction': 0.6805403325957904, 'reg__bagging_fraction': 0.5776566159609288, 'reg__bagging_freq': 6, 'reg__max_bin': 63, 'reg__min_data_per_group': 185}

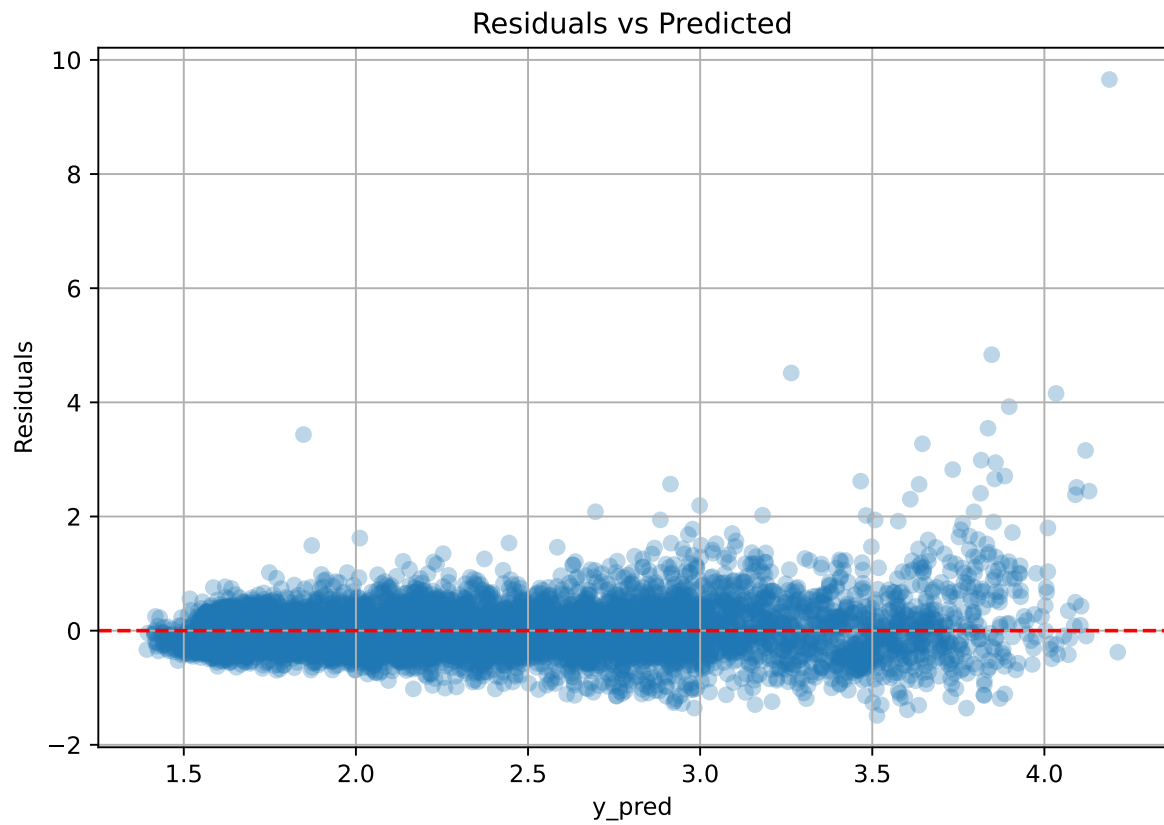
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.042483746862710677, 'reg__num_leaves': 26, 'reg__subsample': 0.8552264814413874, 'reg__colsample_bytree': 0.7958232912019214, 'reg__min_child_samples': 30, 'reg__reg_alpha': 2.112479024076277, 'reg__reg_lambda': 40.614497858467466, 'reg__min_split_gain': 2.2387635879789007, 'reg__path_smooth': 4.153635447090896, 'reg__min_child_weight': 7.122621871188851, 'reg__feature_fraction': 0.7826986189967352, 'reg__bagging_fraction': 0.5467782696770368, 'reg__bagging_freq': 6, 'reg__max_bin': 82, 'reg__min_data_per_group': 192}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.052519477969270575, 'reg__num_leaves': 37, 'reg__subsample': 0.871401909002911, 'reg__colsample_bytree': 0.812289782316176, 'reg__min_child_samples': 36, 'reg__reg_alpha': 3.1248563893086483, 'reg__reg_lambda': 46.66126784318887, 'reg__min_split_gain': 2.33039233284955, 'reg__path_smooth': 3.9818282811768393, 'reg__min_child_weight': 2.130350949807963, 'reg__feature_fraction': 0.8391724692559025, 'reg__bagging_fraction': 0.6112724645428123, 'reg__bagging_freq': 5, 'reg__max_bin': 74, 'reg__min_data_per_group': 176}

3.5.2 Scatter



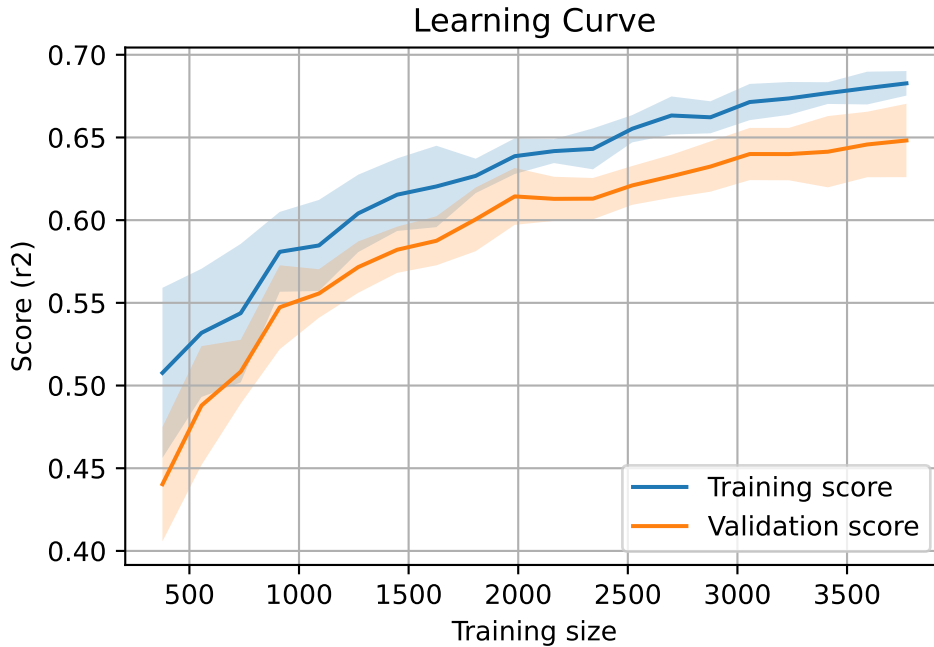
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7698, `Len(training)` : 5389,

`Len(training_real)` : 3773, `Len(test_of_training)` : 1616, `Len(test)` : 2309



3.6 Model : CatBoost

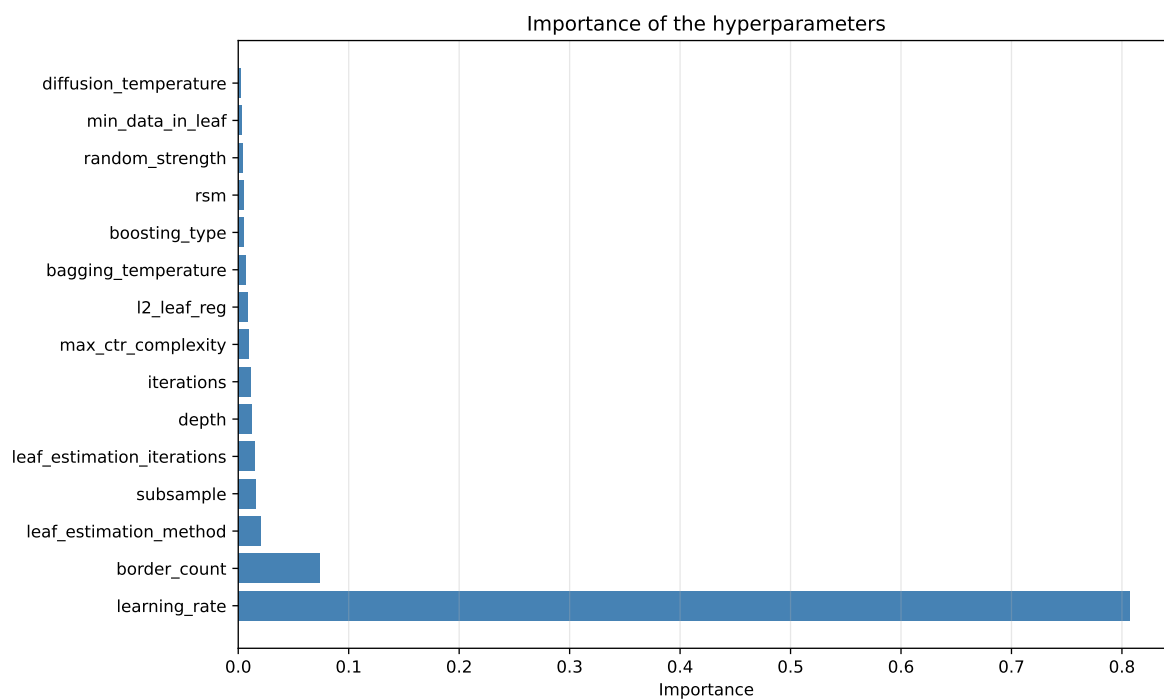
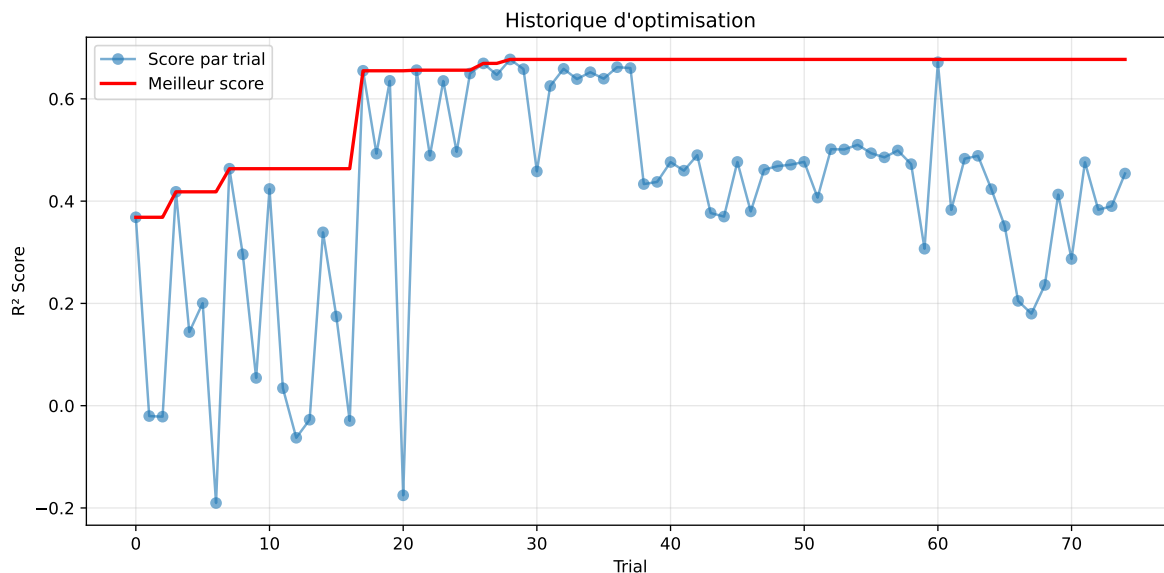
3.6.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.347, std=0.812, min=0.959, max=13.845
 y_{test} : mean=2.347, std=0.772, min=1.011, max=7.779

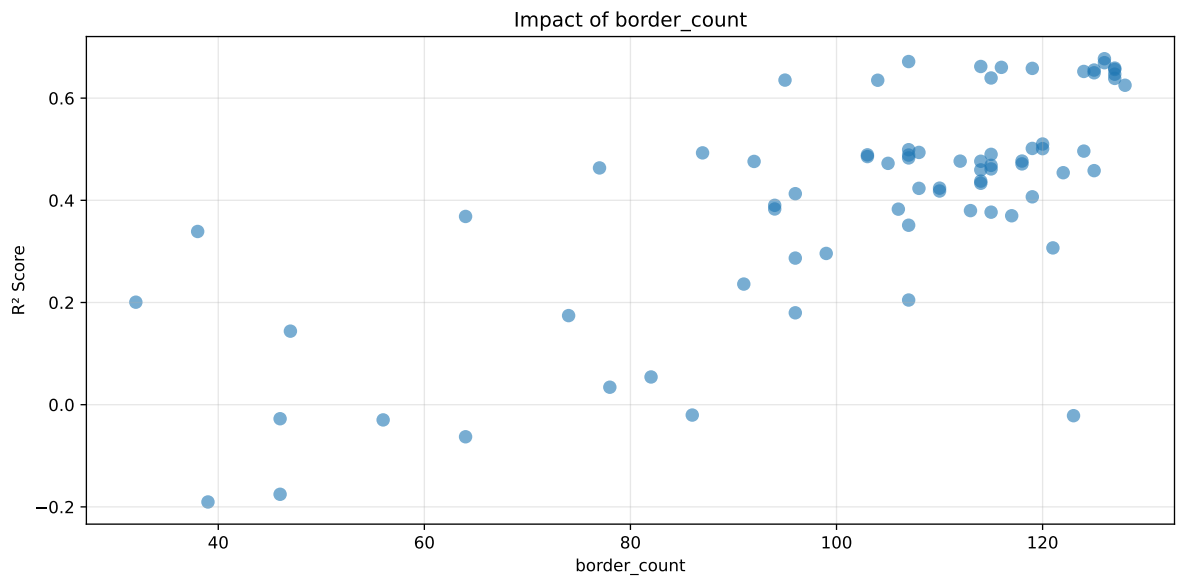
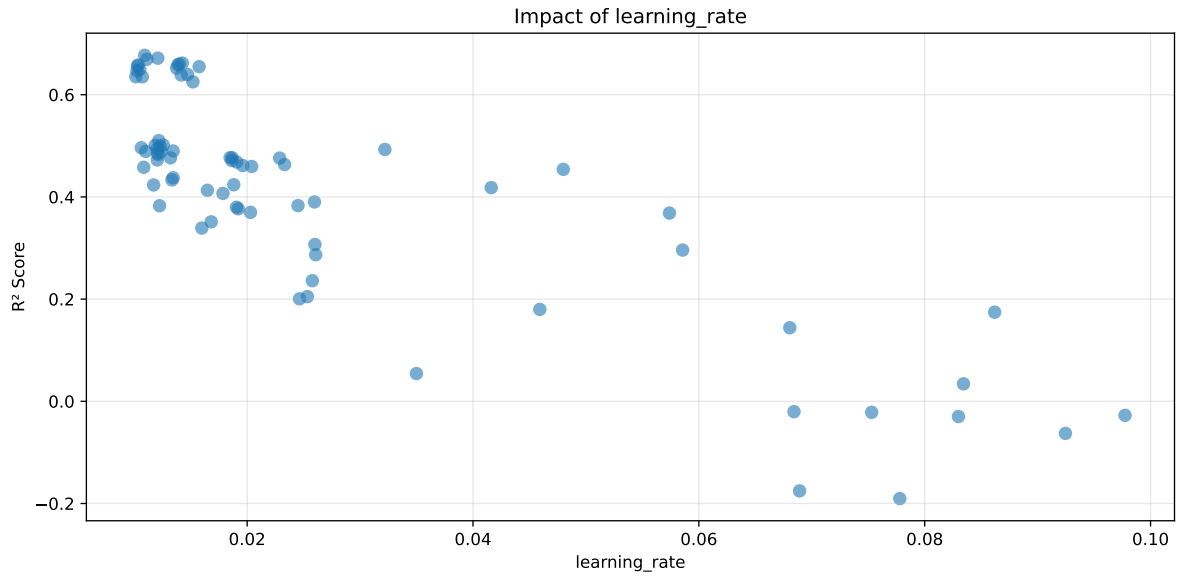
===== TOP Importance FEATURES =====

feature importance	feature	feature importance	feature
21.452690	r_umidity	13.753863	r_temperature_wl_12h_A_energy
9.670042	r_temperature_wl_8h_A_energy_rel	6.935141	r_temperature
5.656822	r_CO2	3.999223	r_temperature_wl_12h_A_energy_rel
4.291081	r_temperature_wl_12h_A_en-	3.377945	r_umidity_wl_12h_A_rms
3.999223	ergy_rel	2.750327	r_THI
3.377945	r_umidity_wl_12h_A_rms	1.918147	r_NH3
2.750327	r_THI	1.820031	r_umidity_wl_8h_A_energy
1.918147	r_NH3	1.782324	r_NH3_wl_8h_D1_energy
1.820031	r_umidity_wl_8h_A_energy	1.777973	r_NH3_wl_12h_D2_energy
1.782324	r_NH3_wl_8h_D1_energy	1.653774	r_NH3_wl_12h_D3_energy
1.777973	r_NH3_wl_12h_D2_energy	1.631662	r_NH3_wl_12h_D1_en-
1.653774	r_NH3_wl_12h_D3_energy	1.518334	r_NH3_wl_8h_D3_energy
1.631662	r_NH3_wl_8h_D3_energy	1.449623	r_umidity_wl_12h_A_energy
1.518334	r_umidity_wl_12h_A_energy	1.404155	r_NH3_wl_8h_A_energy
1.449623	r_NH3_wl_8h_A_energy	1.372916	r_NH3_wl_12h_A_energy
1.404155	r_NH3_wl_12h_A_energy	1.355968	r_umidity_wl_8h_A_energy_rel
1.372916	r_umidity_wl_8h_A_energy_rel	1.214389	r_NH3_wl_12h_D3_energy_rel
1.355968	r_NH3_wl_12h_D3_energy_rel	1.206018	r_NH3_wl_8h_D3_energy_rel
1.214389	r_NH3_wl_8h_D3_energy_rel	1.180079	r_NH3_wl_8h_D2_energy
1.206018	r_NH3_wl_8h_D2_energy	0.843575	r_temperature_wl_8h_D3_energy
1.180079	r_temperature_wl_8h_D3_energy	0.837727	r_NH3_wl_12h_A_rms
0.843575	r_NH3_wl_12h_A_rms	0.770399	r_umidity_wl_12h_D3_energy
0.837727	r_umidity_wl_12h_D3_energy	0.649609	r_umidity_wl_8h_D2_energy_rel
0.770399	r_umidity_wl_8h_D2_energy_rel	0.561903	r_NH3_wl_12h_A_en-
0.649609	r_NH3_wl_12h_A_en-	0.535188	ergy_rel
0.561903	ergy_rel	0.512868	r_umidity_wl_8h_A_rms
0.535188	r_umidity_wl_8h_A_rms	0.506821	r_NH3_wl_8h_A_energy_rel
0.512868	r_NH3_wl_8h_A_energy_rel	0.481361	r_umidity_wl_8h_D3_energy_rel
0.506821	r_umidity_wl_8h_D3_energy_rel		
0.481361			

r_temperature_wl_12h_D3_energy 0.408402 r_temperature_wl_8h_A_rms 0.400257
r_umidity_wl_12h_D2_energy_rel 0.319363

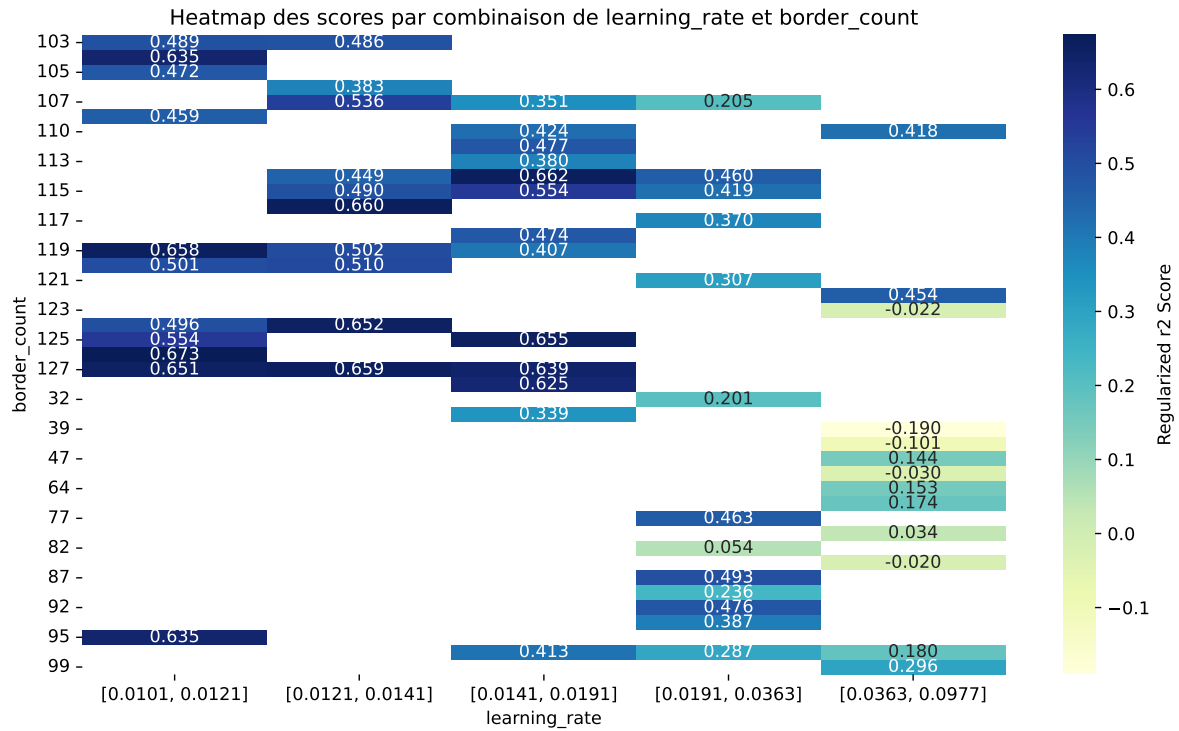


Paramètres avec >5% d'importance : ['learning_rate', 'border_count']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7091	0.6771	0.6826	0.0183	0.6771	0.6951	0.0179	1.0265	1.0473
2	0.7018	0.6716	0.6781	0.0182	0.6716	0.6876	0.016	1.0238	1.0451
3	0.6972	0.6694	0.673	0.019	0.6694	0.6885	0.0192	1.0286	1.0417
4	0.695	0.6619	0.6709	0.0186	0.6619	0.681	0.0191	1.0289	1.05
5	0.6925	0.6602	0.6664	0.0191	0.6602	0.6782	0.0179	1.0272	1.0488
6	0.684	0.6588	0.6617	0.016	0.6588	0.6737	0.015	1.0227	1.0383
7	0.684	0.6582	0.6622	0.0185	0.6582	0.6749	0.0167	1.0253	1.0392
8	0.6802	0.6561	0.659	0.0164	0.6561	0.6683	0.0122	1.0186	1.0368
9	0.6814	0.6549	0.6606	0.0168	0.6549	0.6673	0.0124	1.019	1.0404
10	0.6782	0.6522	0.6546	0.0169	0.6522	0.6653	0.0131	1.0201	1.0398
---	---	---	---	---	---	---	---	---	---
75	0.9954	0.7978	0.8222	0.0281	-0.1904	0.8343	0.0365	1.0457	1.2477

Hyperparameters:

Rank 1: {'reg__iterations': 1000, 'reg__depth': 4, 'reg__learning_rate': 0.010929273572599396, 'reg__l2_leaf_reg': 33.736914868187, 'reg__bagging_temperature': 1.2822794947697789, 'reg__random_strength': 1.3671842537786199, 'reg__subsample': 0.6935872600407036, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5669822187158077, 'reg__border_count': 126, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 547.2828554091199}

Rank 2: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.012088722260231574, 'reg__l2_leaf_reg': 34.079649617898404, 'reg__bagging_temperature': 2.6823563076328085, 'reg__random_strength': 1.7949358762268317, 'reg__subsample': 0.5473529076317176, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5434683344588056, 'reg__border_count': 107, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2816.512031768594}

Rank 3: {'reg__iterations': 1000, 'reg__depth': 4, 'reg__learning_rate': 0.011124543077200004, 'reg__l2_leaf_reg': 49.54108151456254, 'reg__bagging_temperature': 2.7872336964310014, 'reg__random_strength': 1.441661344253506, 'reg__subsample': 0.6714189376062036, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5702517835944741, 'reg__border_count': 126, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 369.63918355387875}

Rank 4: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.01426406924828991, 'reg__l2_leaf_reg': 44.94685214323225, 'reg__bagging_temperature': 2.414759602242766, 'reg__random_strength': 4.673255291061316, 'reg__subsample': 0.7415225350892938, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5559334154662513, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1525.4467856821107}

Rank 5: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.013990880074614104, 'reg__l2_leaf_reg': 43.433943483957556, 'reg__bagging_temperature': 2.432118154695611, 'reg__random_strength': 4.591262091934295, 'reg__subsample': 0.7526446745789366, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.548784373479232, 'reg__border_count': 116, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1411.6113540039983}

Rank 6: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.013867768212517994, 'reg__l2_leaf_reg': 36.19606209990208, 'reg__bagging_temperature': 2.4273178918084595, 'reg__random_strength': 1.3490819365892737, 'reg__subsample': 0.5138848195190374, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5831487374493157, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 164.82292121155479}

Rank 7: {'reg__iterations': 1000, 'reg__depth': 4, 'reg__learning_rate': 0.010349512684734722, 'reg__l2_leaf_reg': 49.83119275432337, 'reg__bagging_temperature': 1.007644654881055, 'reg__random_strength': 1.3608357229843258, 'reg__subsample': 0.7448518038886061,

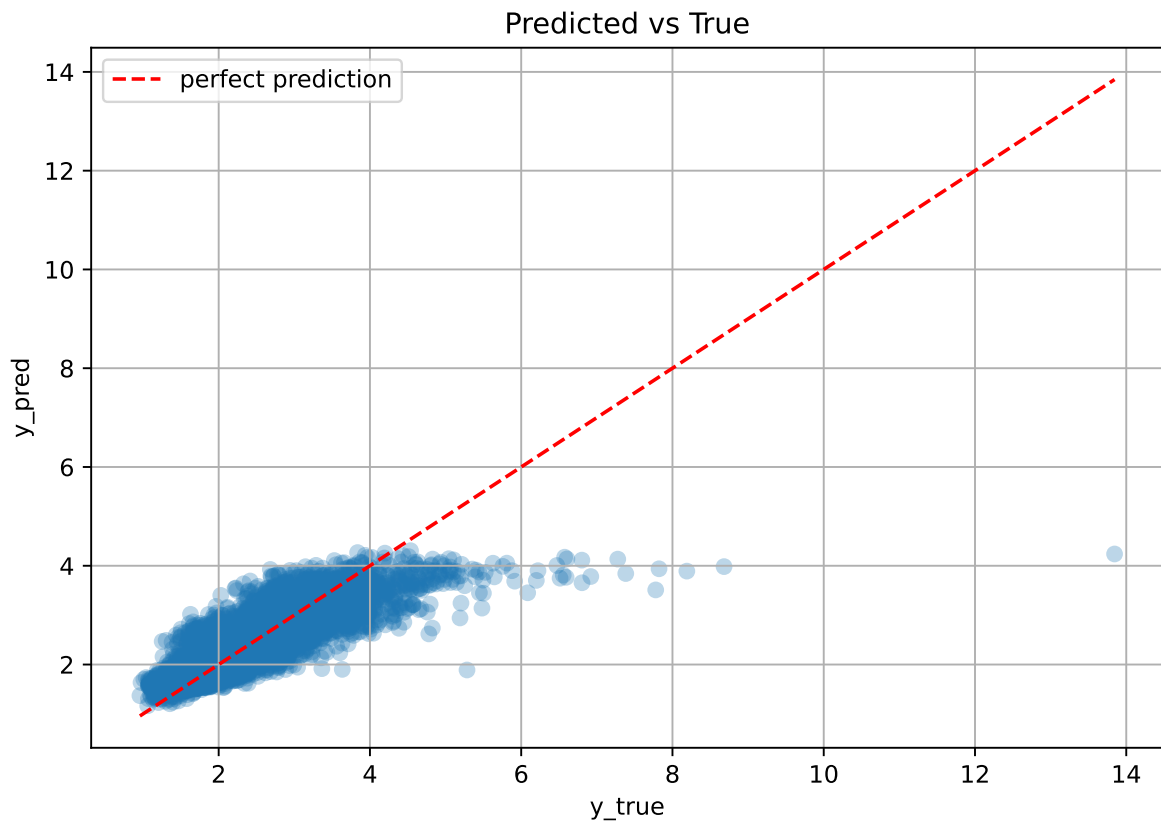
'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5754718448010078, 'reg__border_count': 119, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 192.9598337551073}

Rank 8: {'reg__iterations': 1000, 'reg__depth': 4, 'reg__learning_rate': 0.010289563441882501, 'reg__l2_leaf_reg': 35.985461876414504, 'reg__bagging_temperature': 3.108064996690504, 'reg__random_strength': 3.928243177349755, 'reg__subsample': 0.5102060592362471, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5076578233251597, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 9762.412979521921}

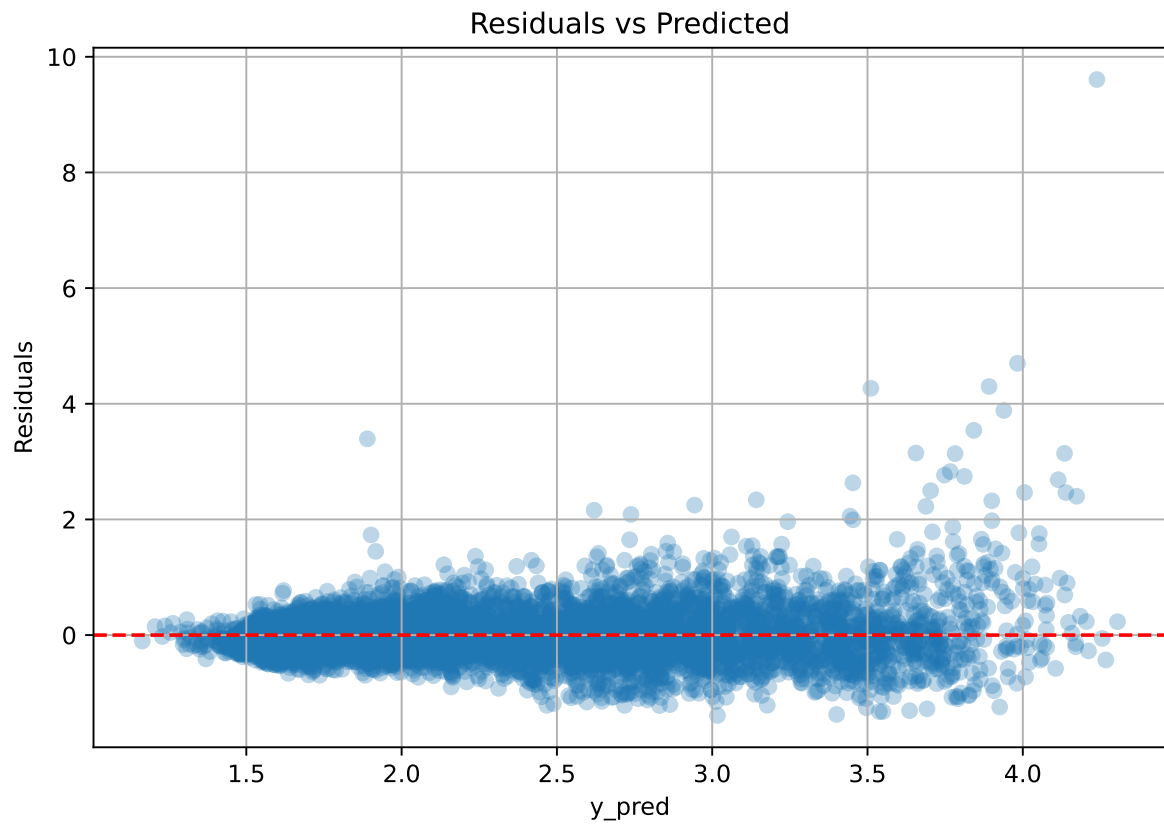
Rank 9: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.01574808972240963, 'reg__l2_leaf_reg': 42.68447031334202, 'reg__bagging_temperature': 2.7566912013510203, 'reg__random_strength': 1.3925534983823744, 'reg__subsample': 0.6552084522170927, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.5631414224439368, 'reg__border_count': 125, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 612.9358291227783}

Rank 10: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.013747598082920151, 'reg__l2_leaf_reg': 48.575068632379256, 'reg__bagging_temperature': 2.335781474191222, 'reg__random_strength': 0.9496842254751265, 'reg__subsample': 0.7444592982947427, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5690772619128187, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 364.8655815259096}

3.6.2 Scatter



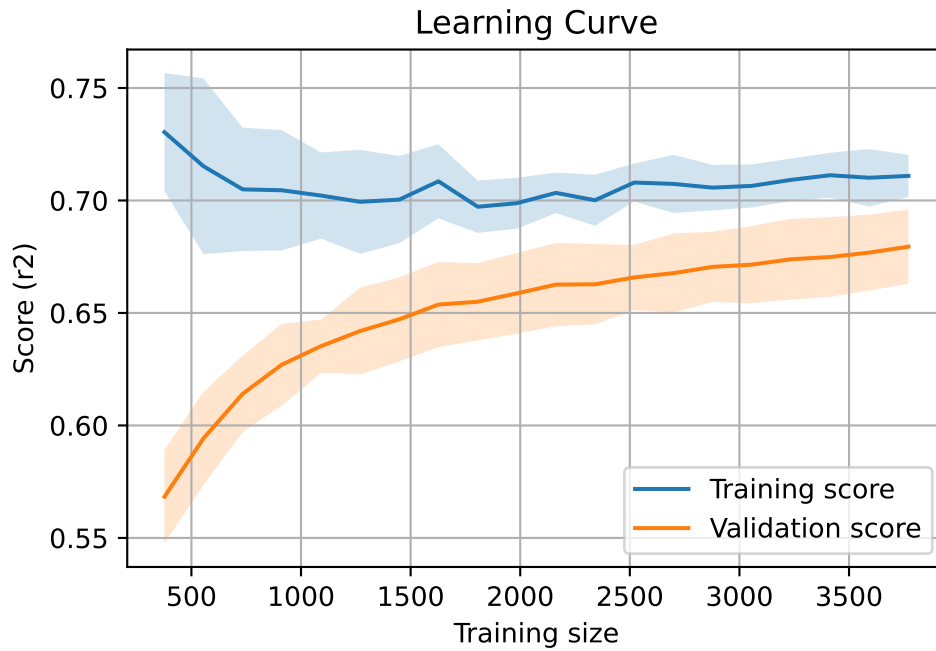
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7698, `Len(training)` : 5389,

`Len(training_real)` : 3773, `Len(test_of_training)` : 1616, `Len(test)` : 2309



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.56	0.551	0.012	0.55	0.55	-0.001	0.998	1.016	nan
LGBM	0.683	0.652	0.021	0.657	0.682	0.03	1.047	1.048	0.6516
CatBoost	0.709	0.677	0.018	0.683	0.695	0.018	1.026	1.047	0.6771

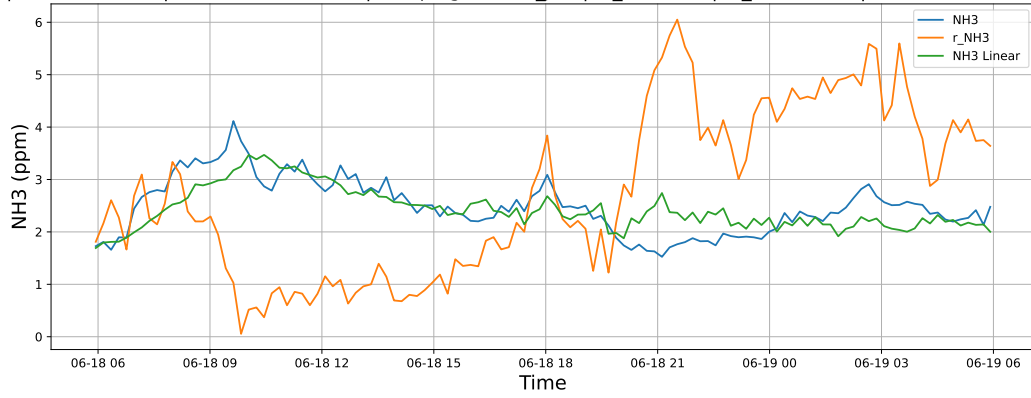
4 Plot

4.1 Standalone plots

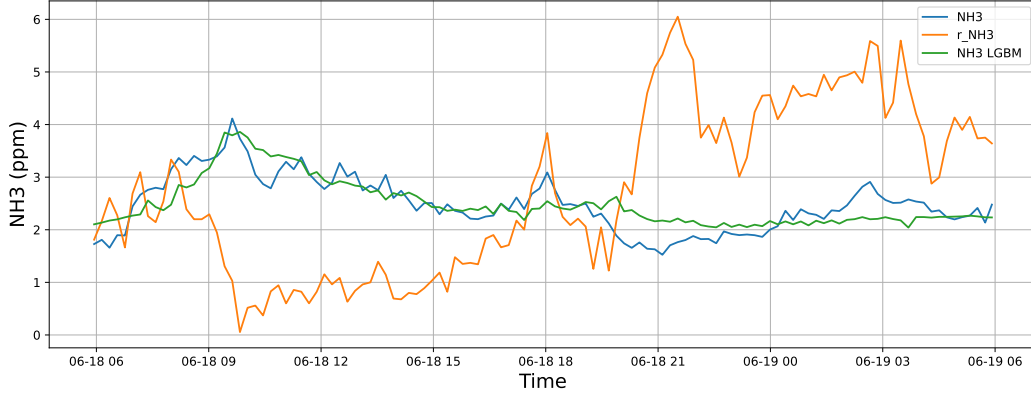
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

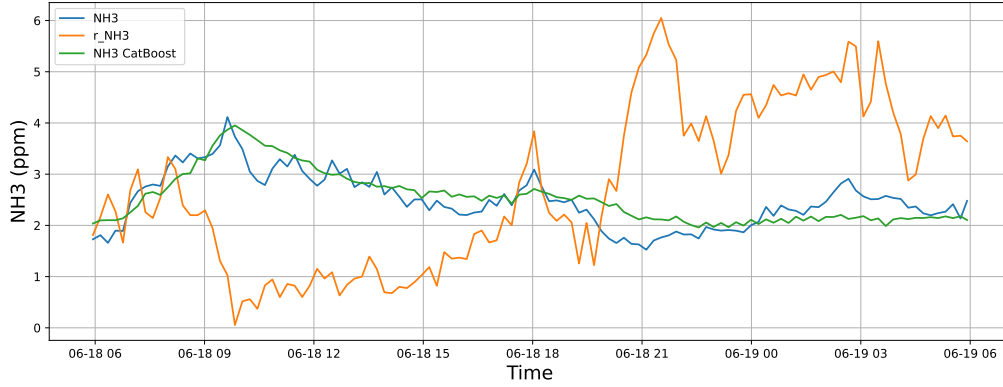
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

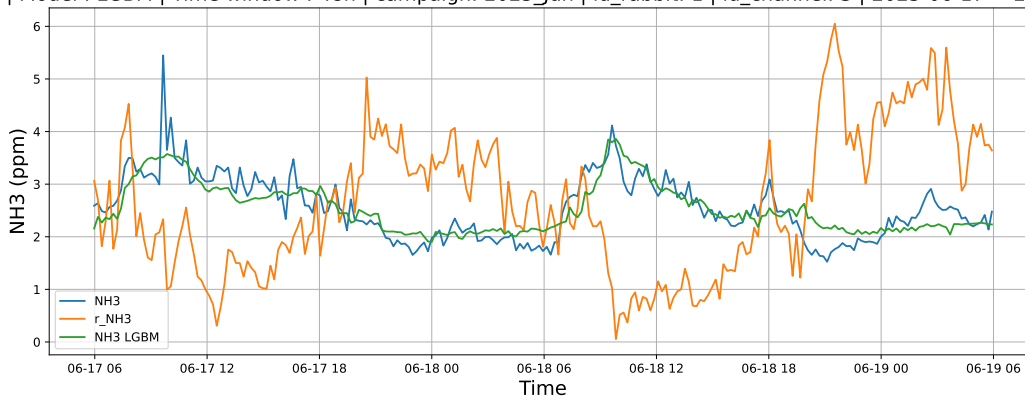


4.1.1.2 Time window : 48

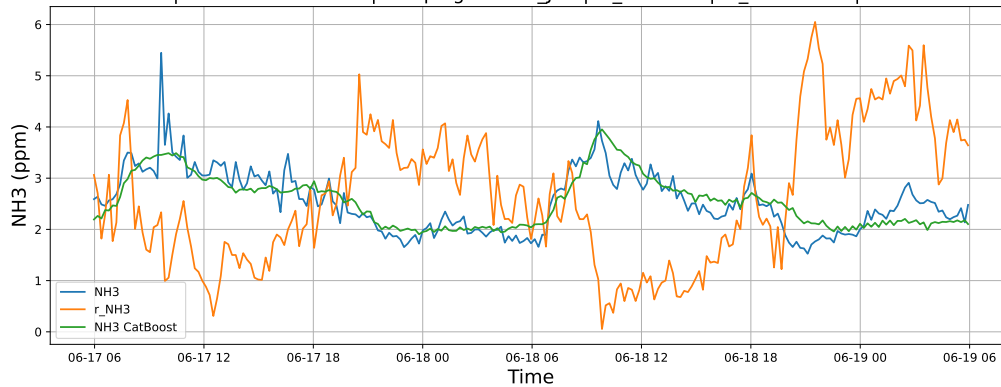
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

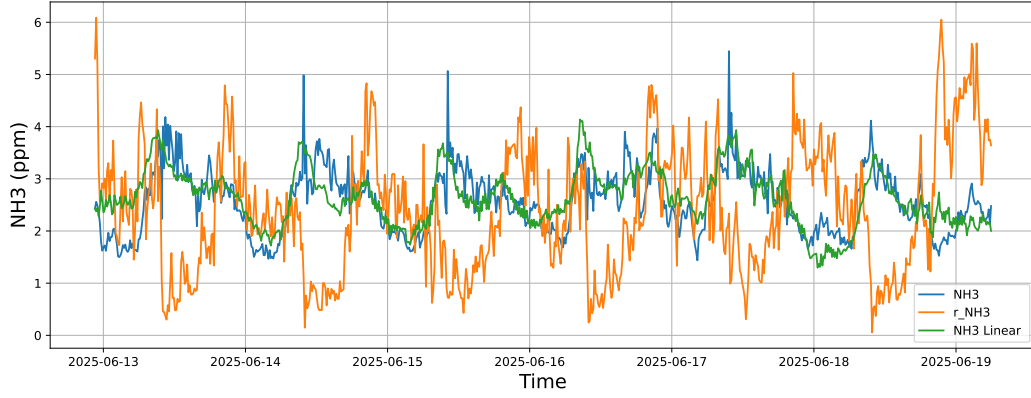


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

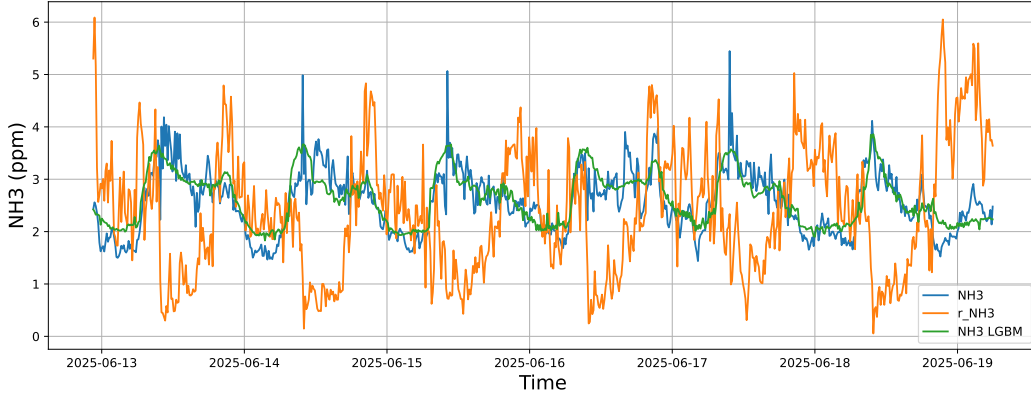


4.1.1.3 Time window : ALL

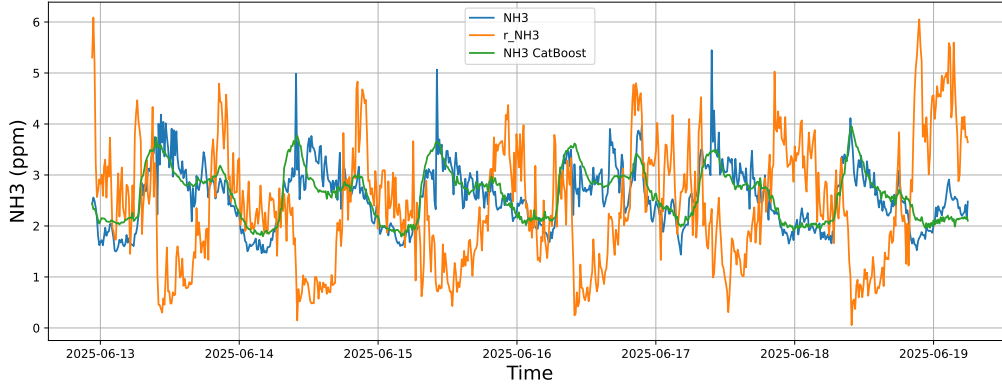
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



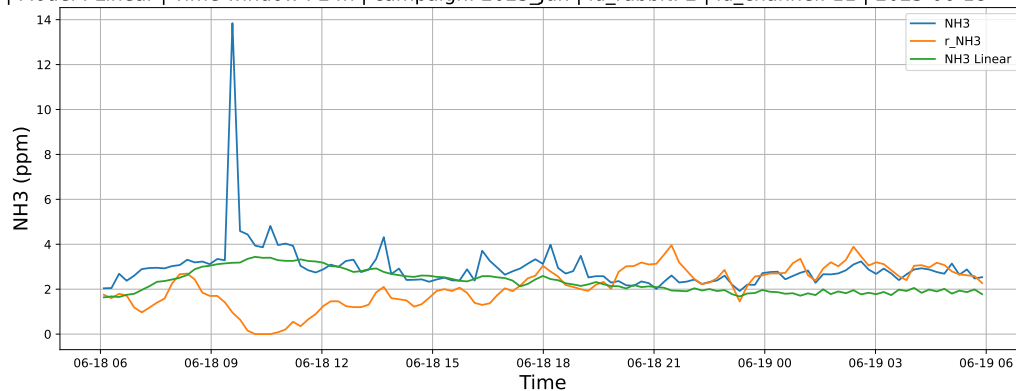
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



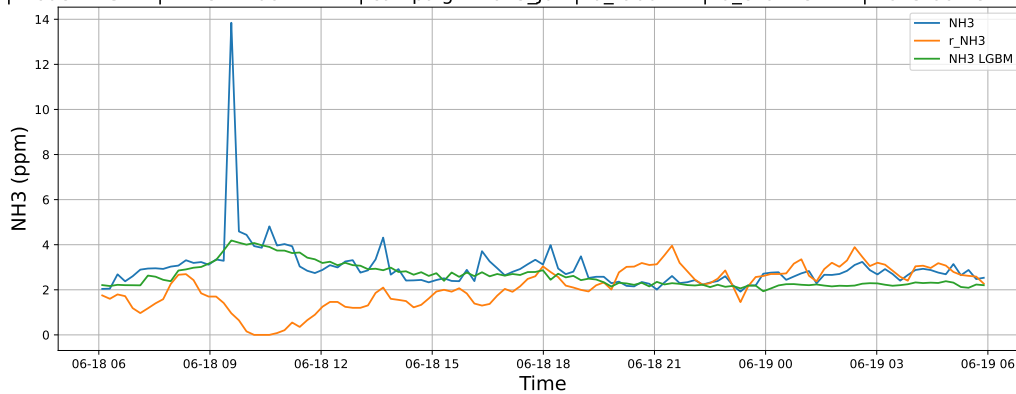
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

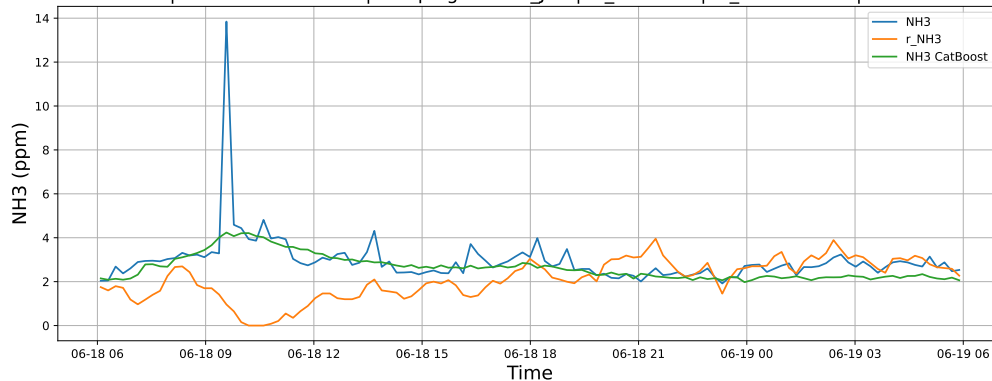
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

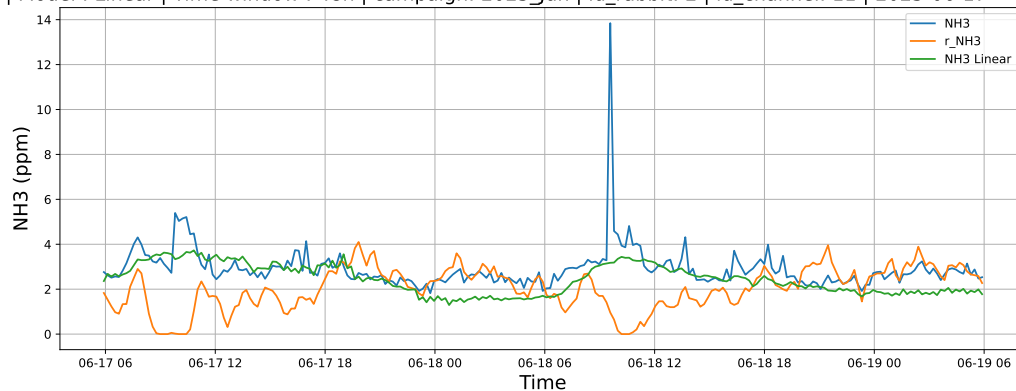


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

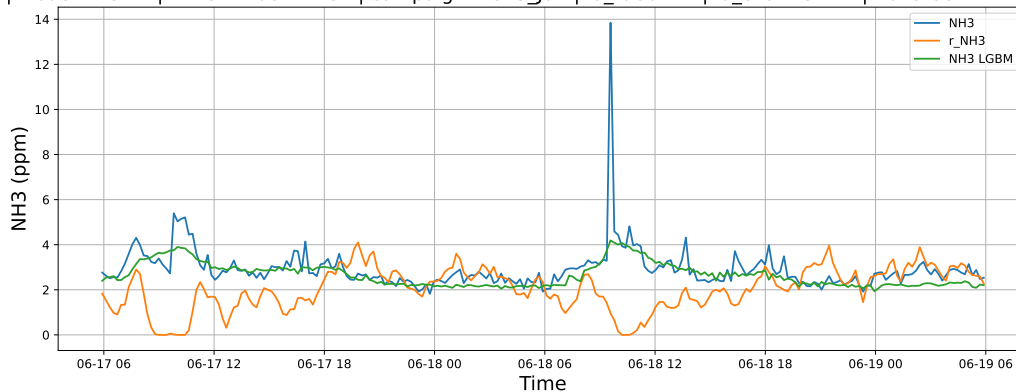


4.1.2.2 Time window : 48

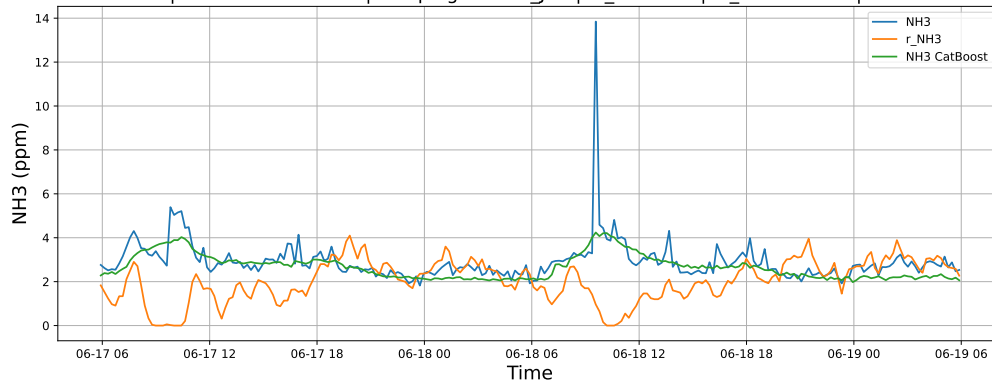
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

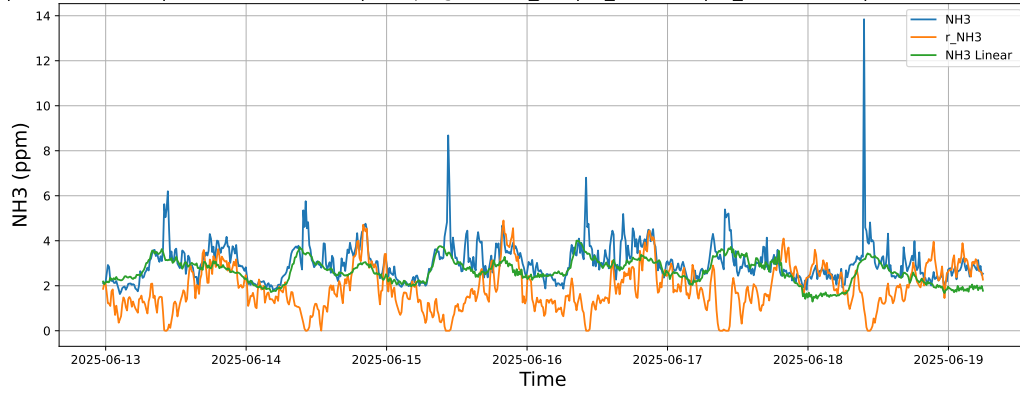


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

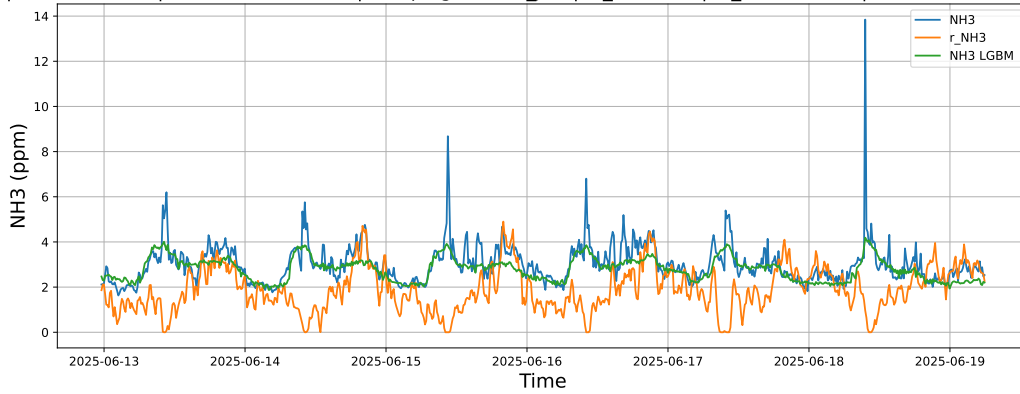


4.1.2.3 Time window : ALL

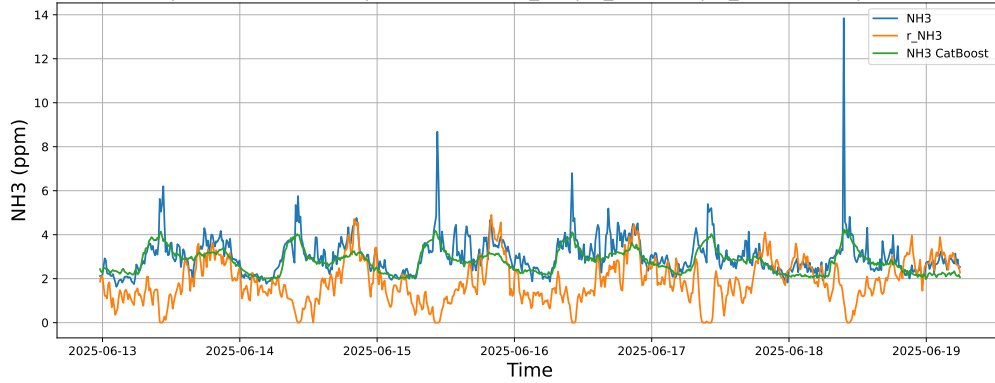
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



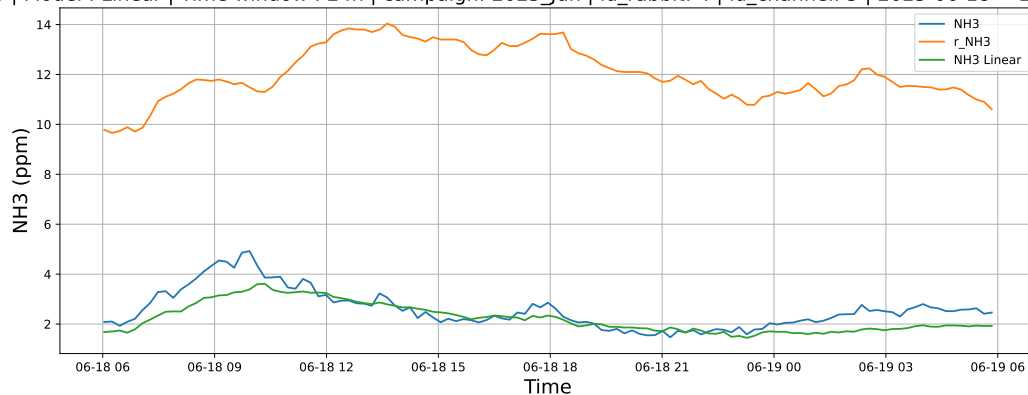
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



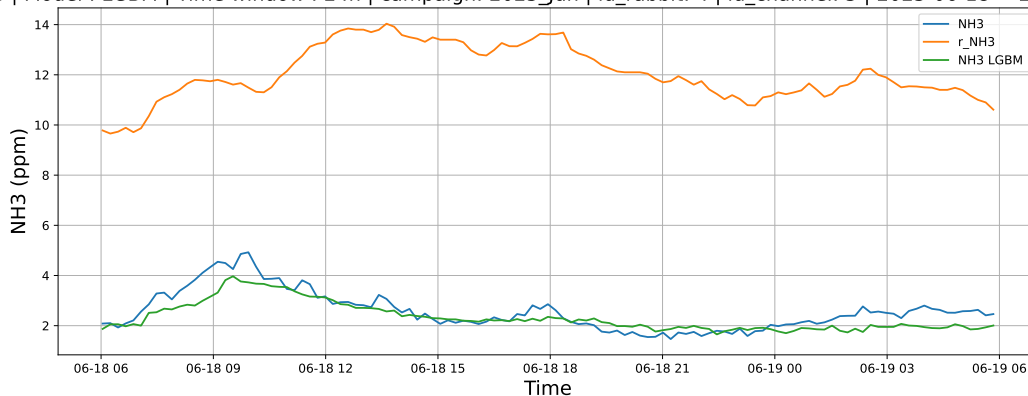
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

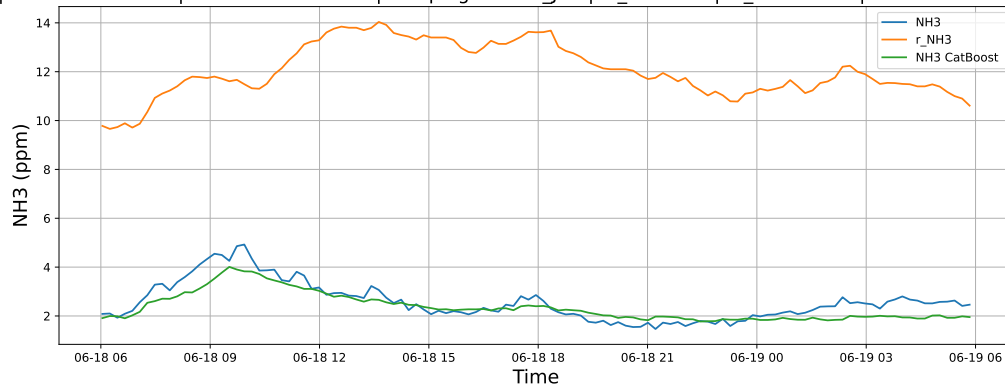
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

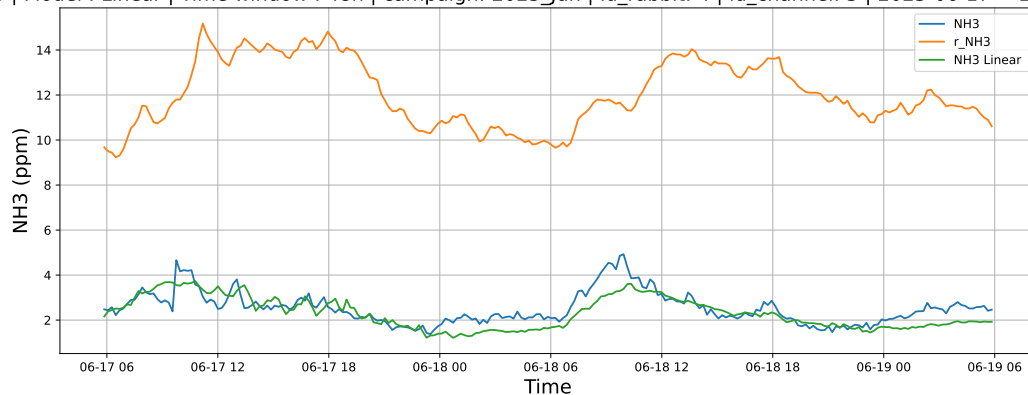


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

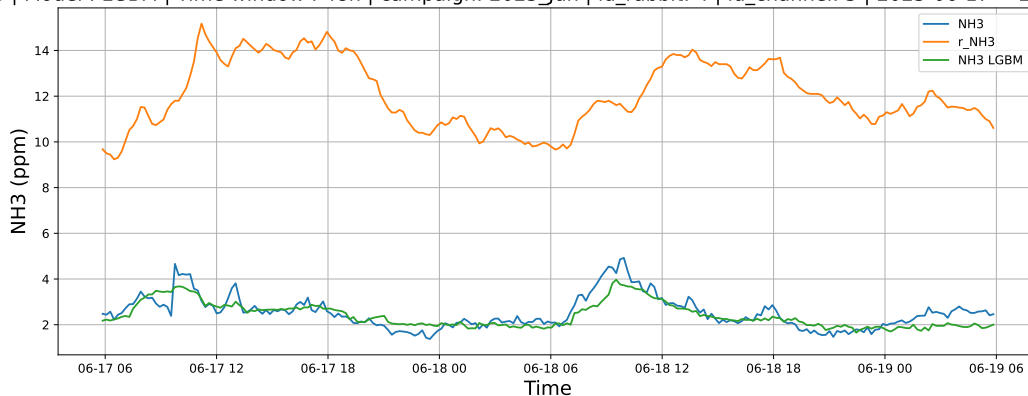


4.1.3.2 Time window : 48

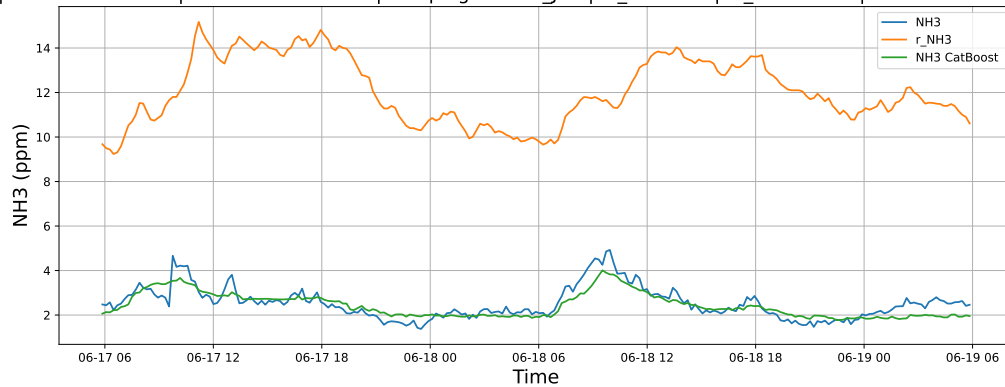
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

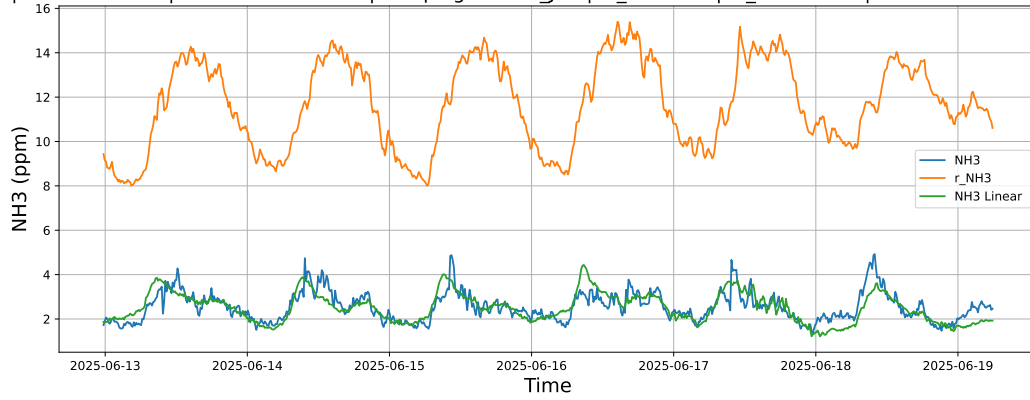


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

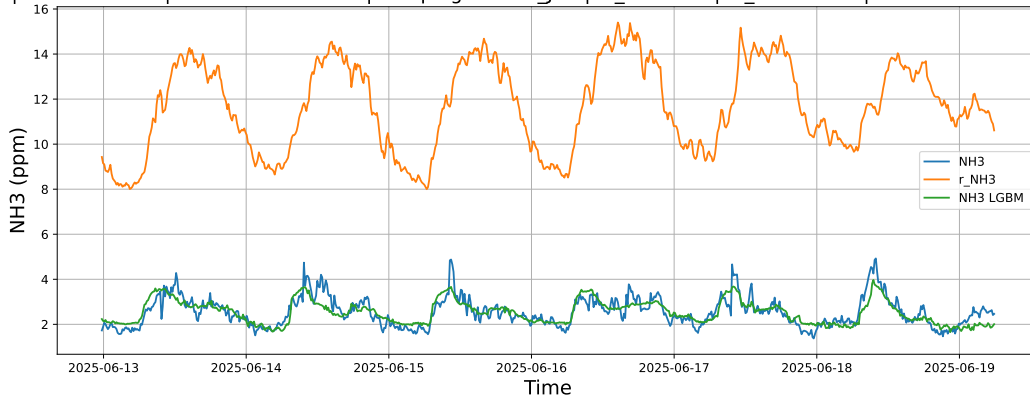


4.1.3.3 Time window : ALL

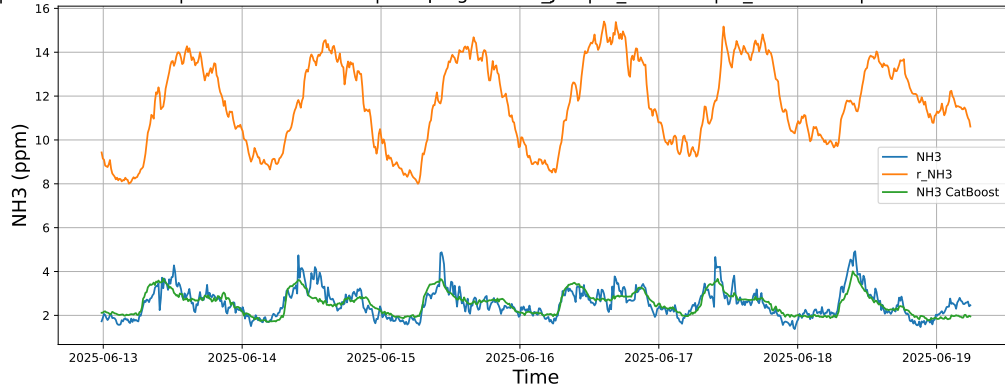
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



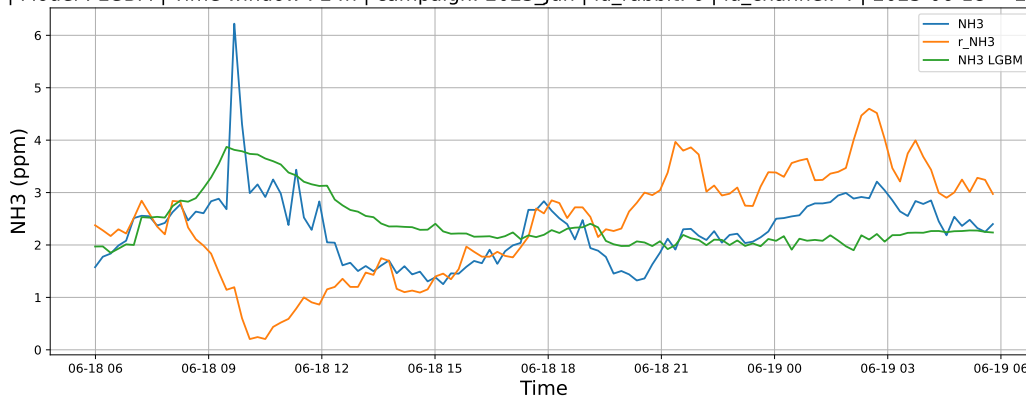
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

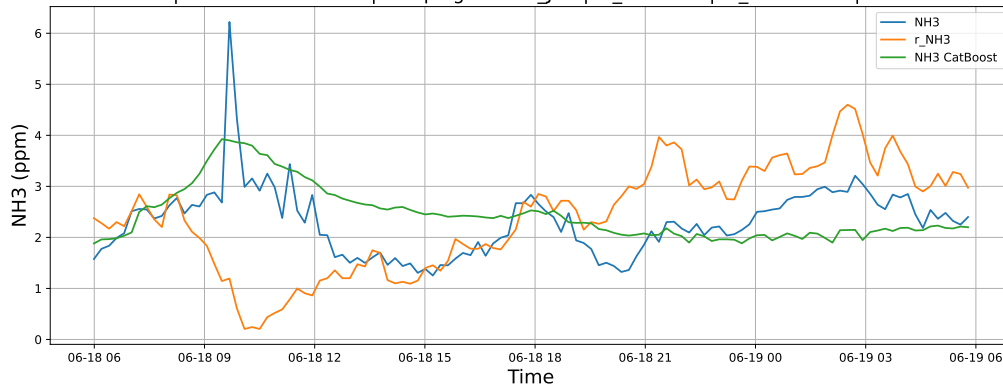
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

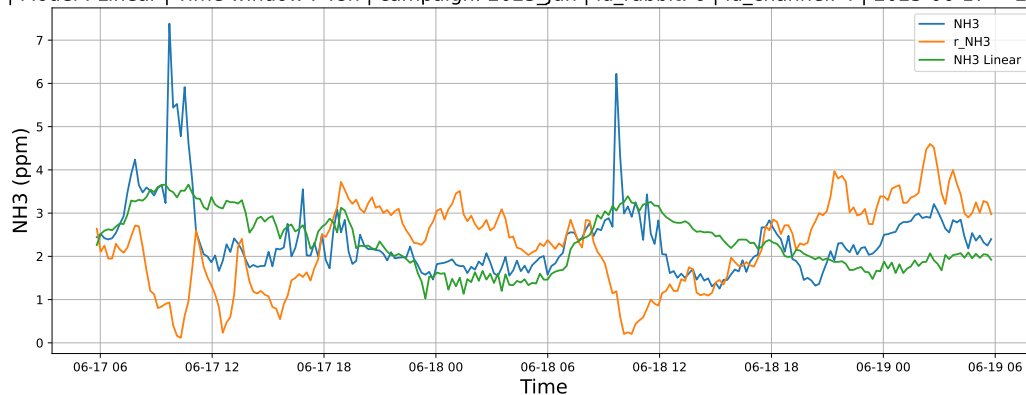


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

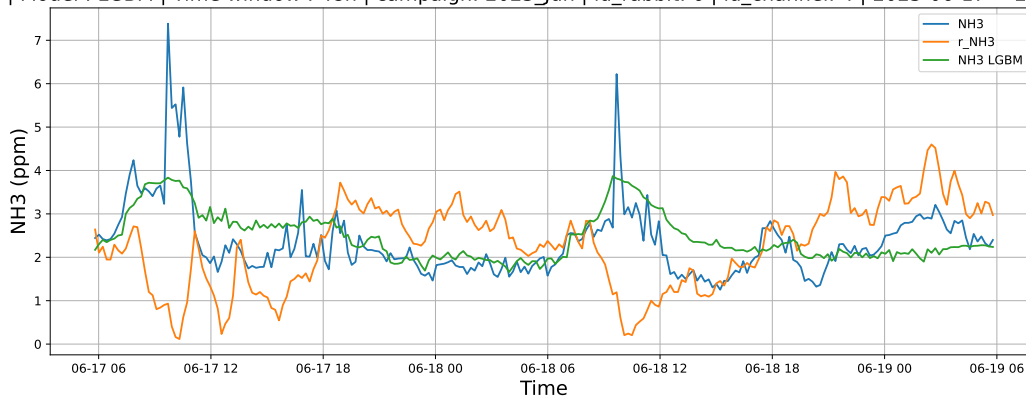


4.1.4.2 Time window : 48

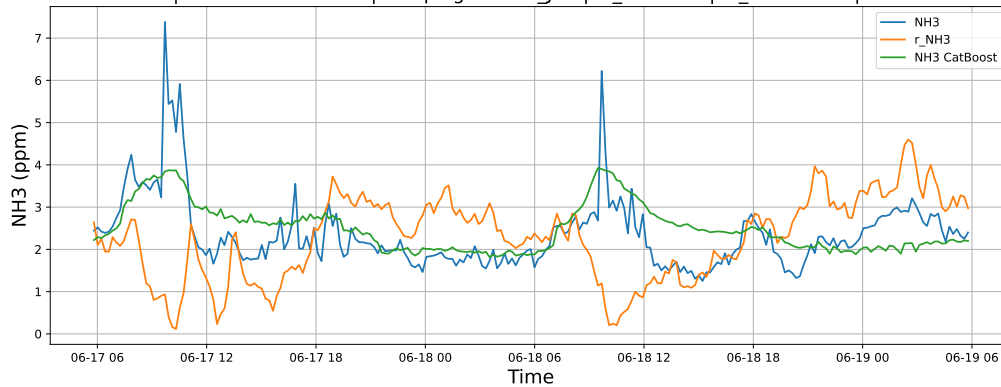
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

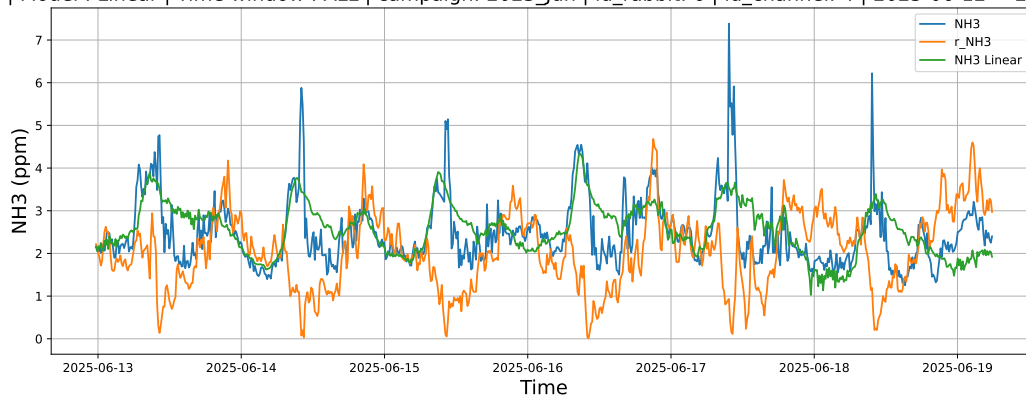


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

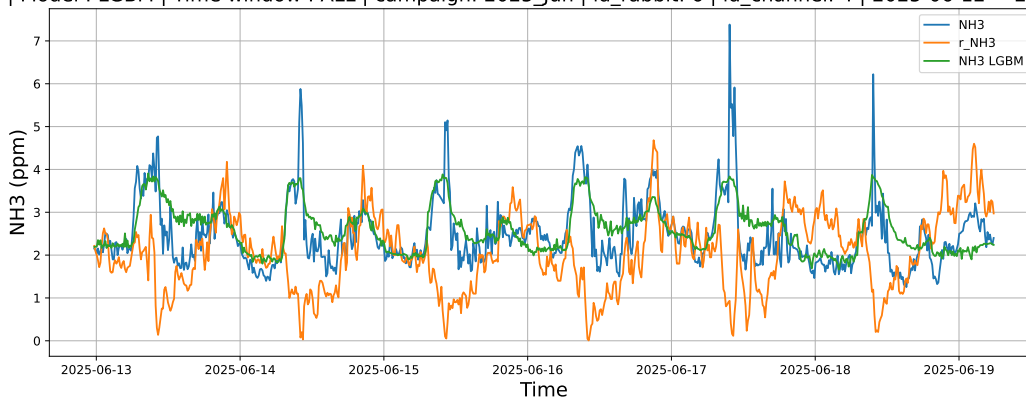


4.1.4.3 Time window : ALL

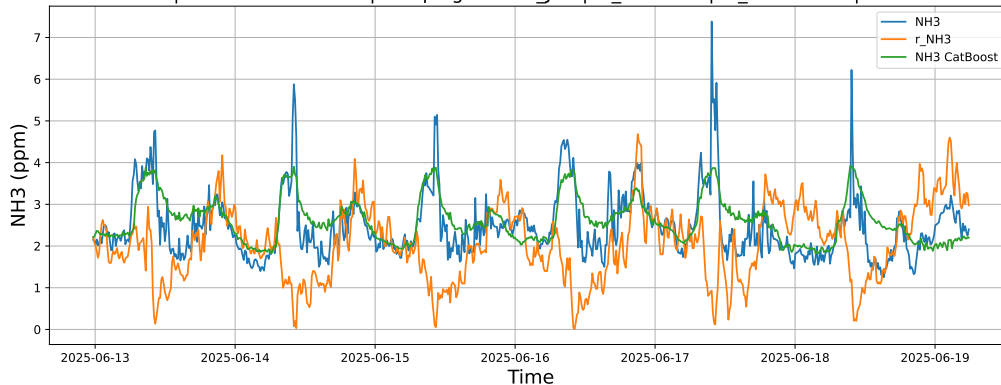
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



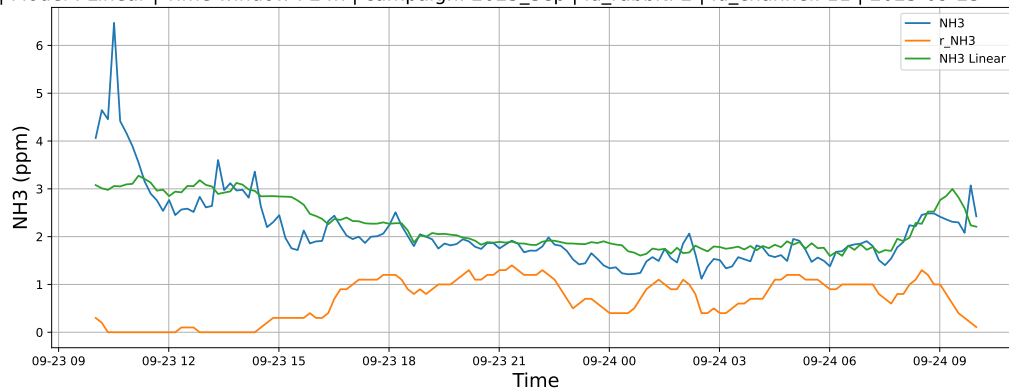
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



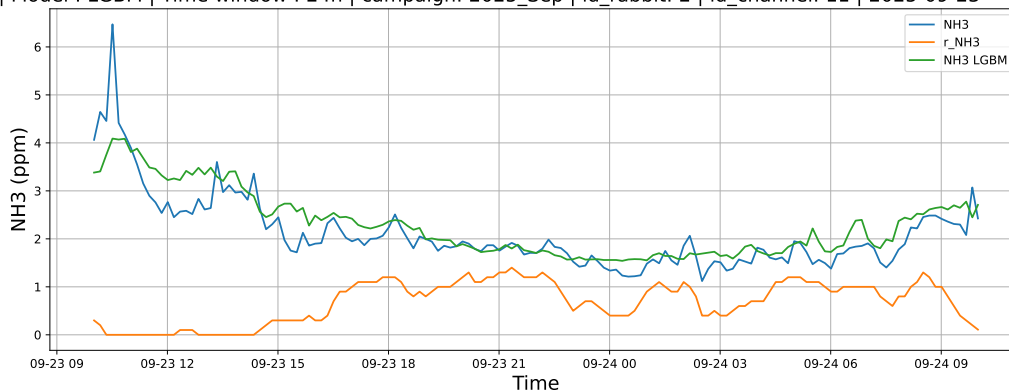
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

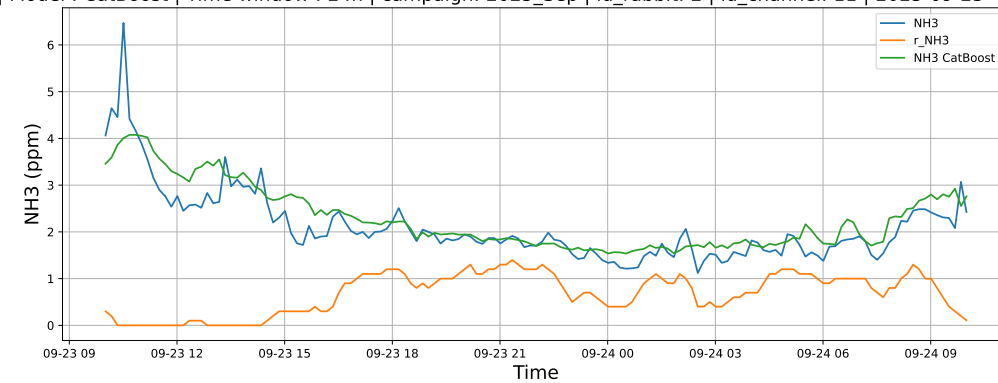
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

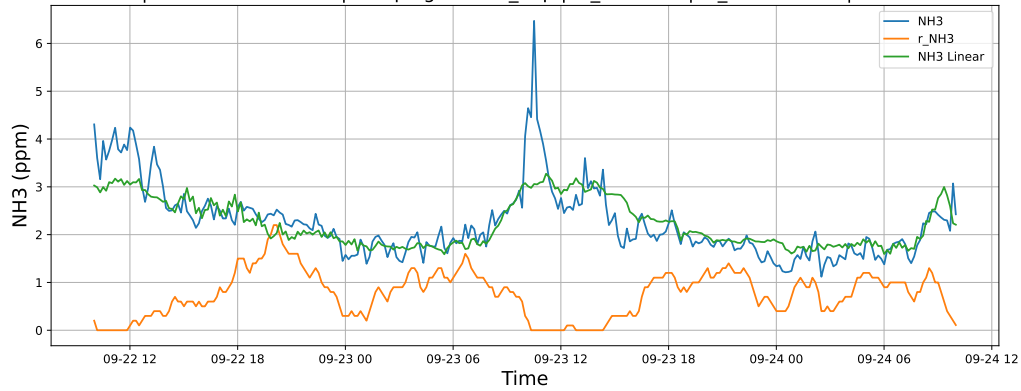


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

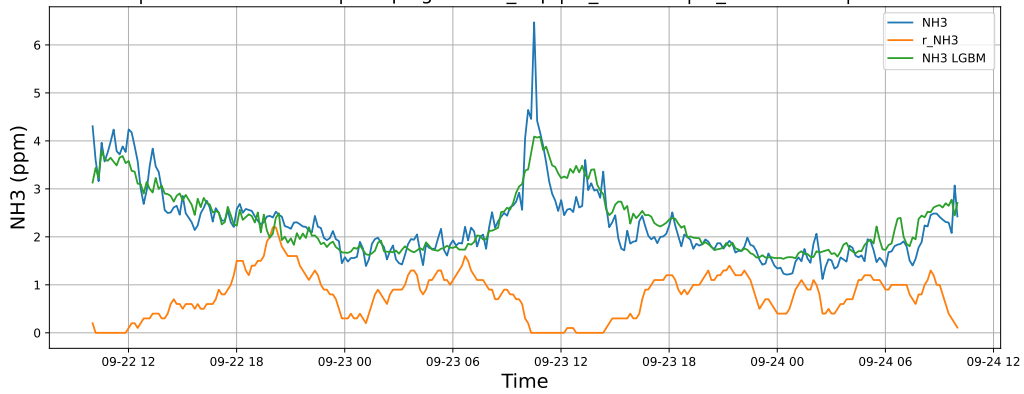


4.1.5.2 Time window : 48

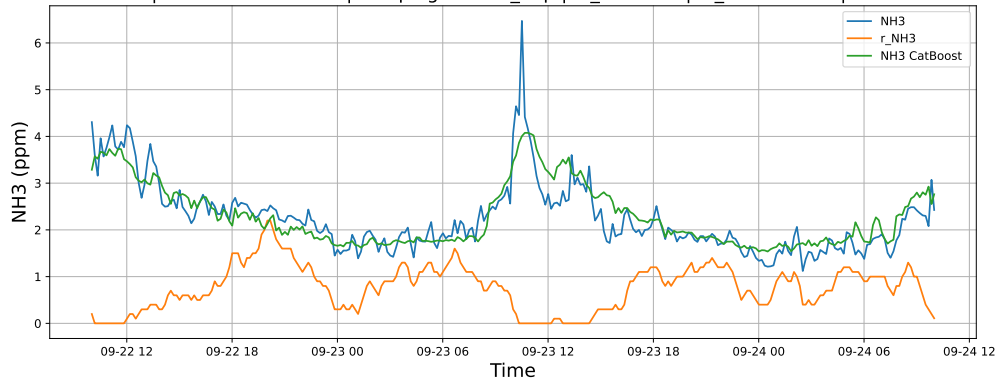
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

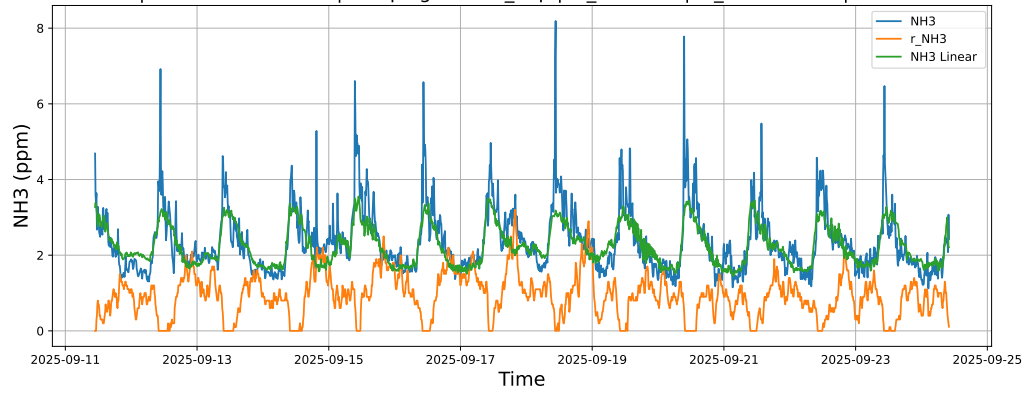


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

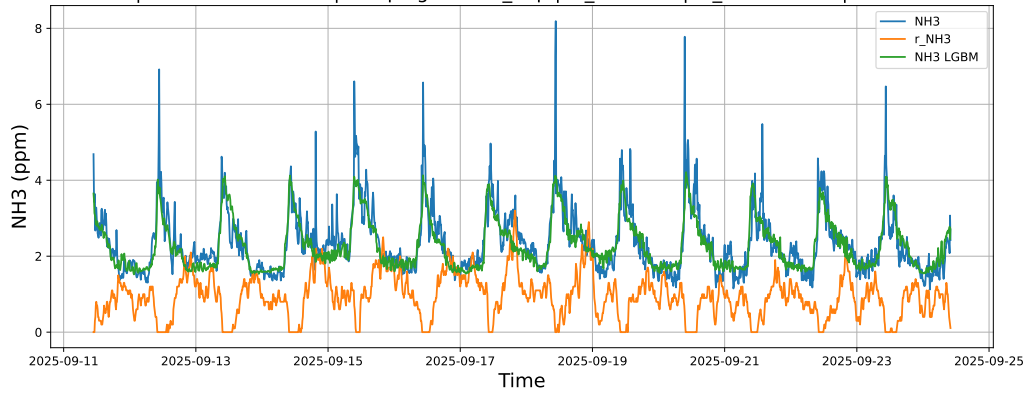


4.1.5.3 Time window : ALL

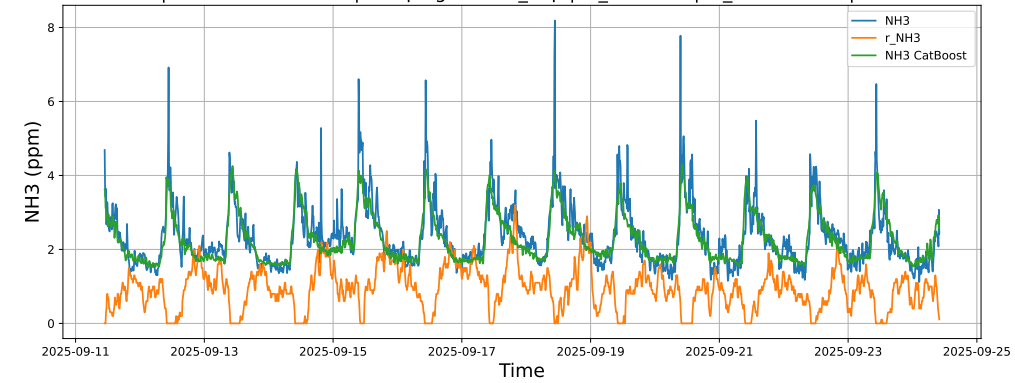
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



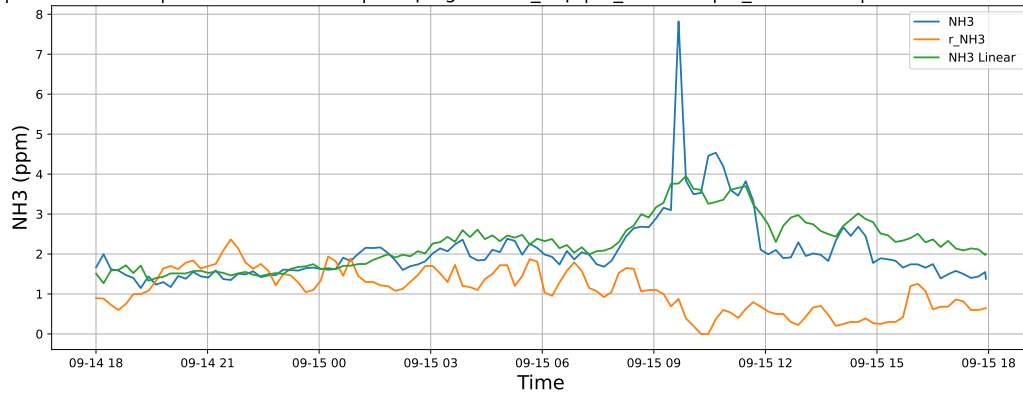
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



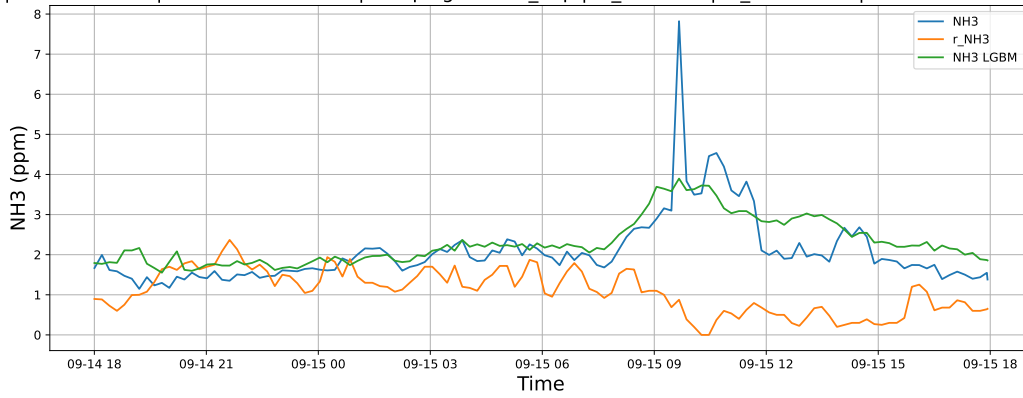
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

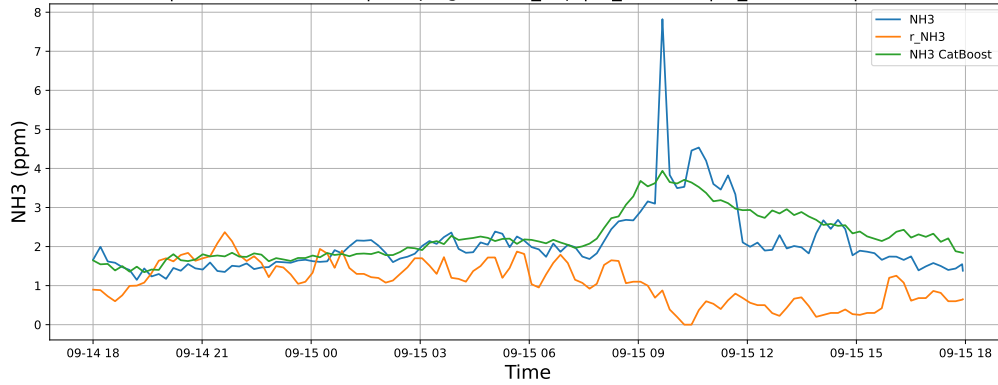
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

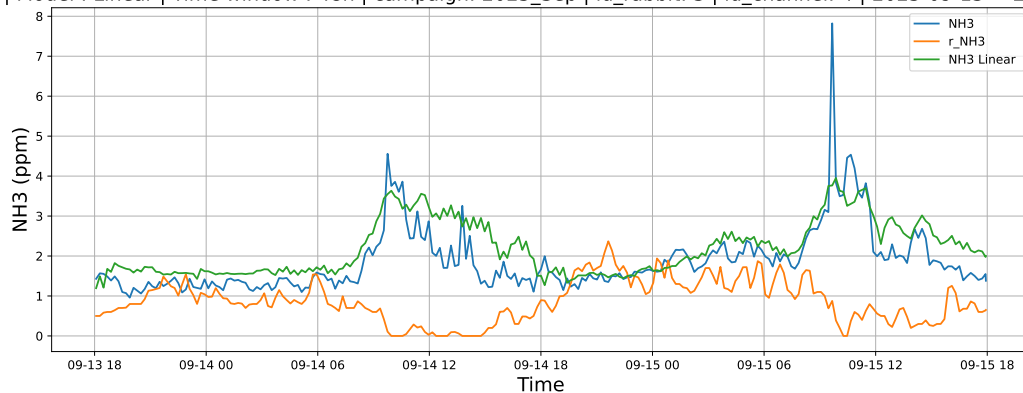


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

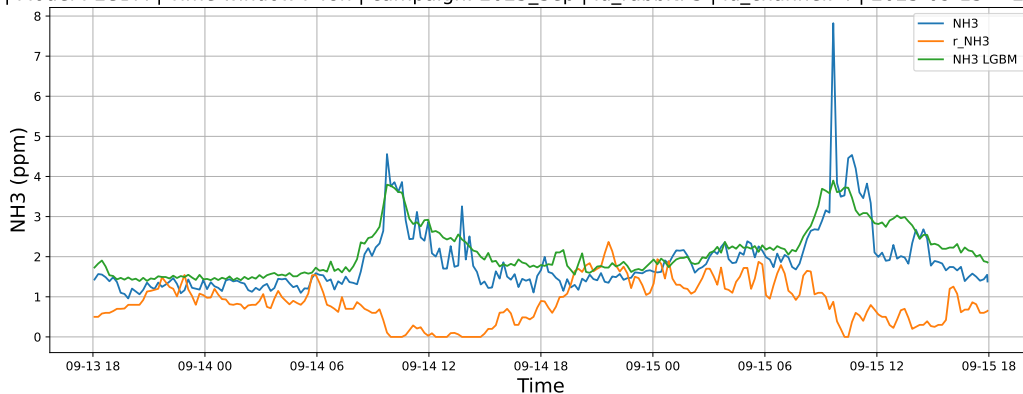


4.1.6.2 Time window : 48

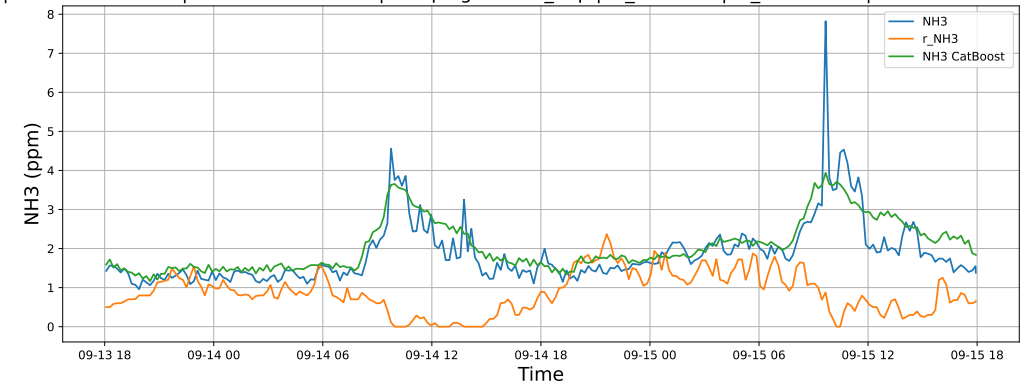
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

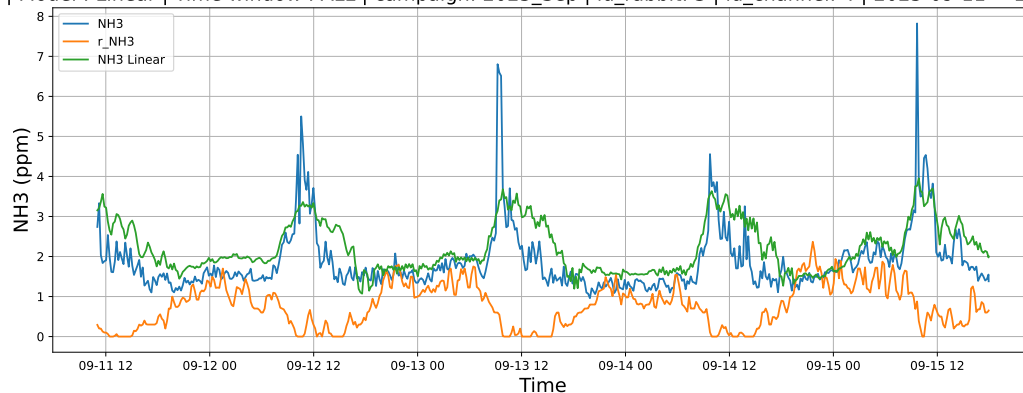


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

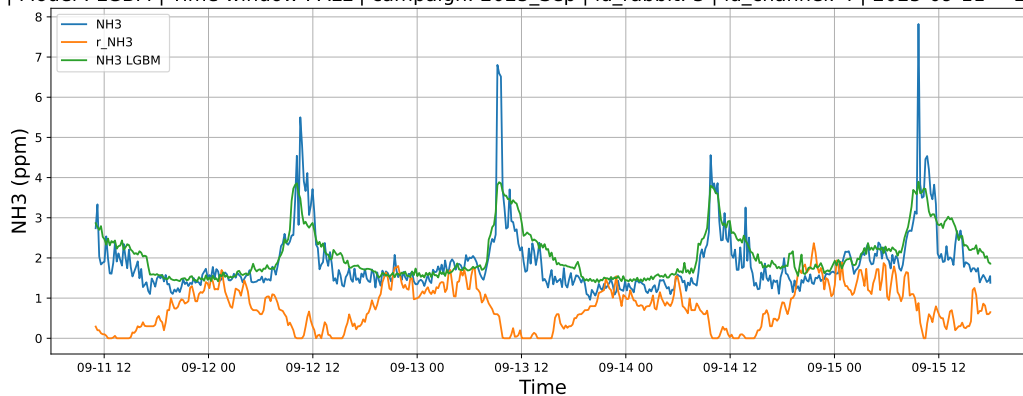


4.1.6.3 Time window : ALL

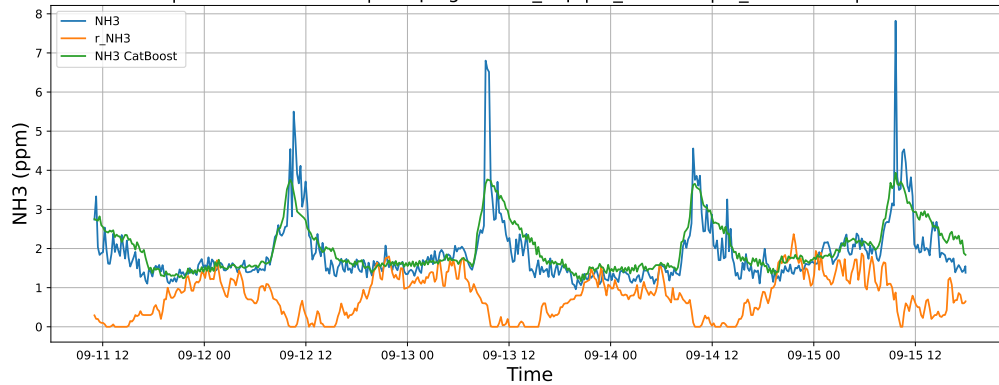
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



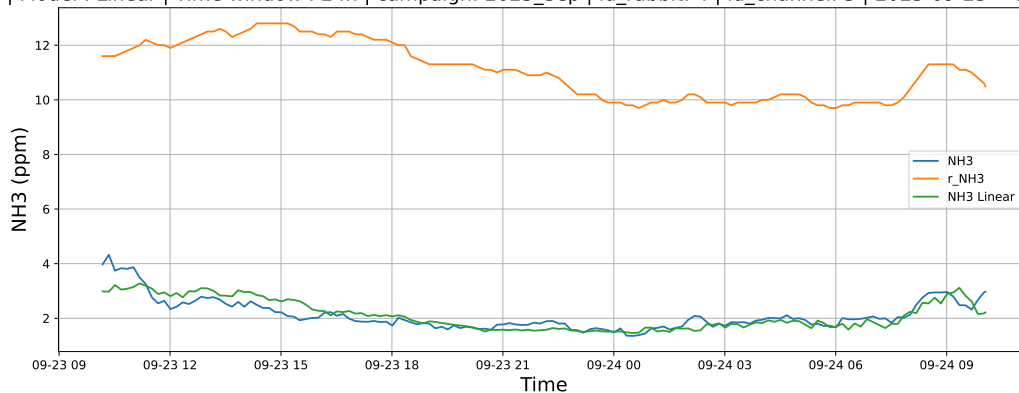
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



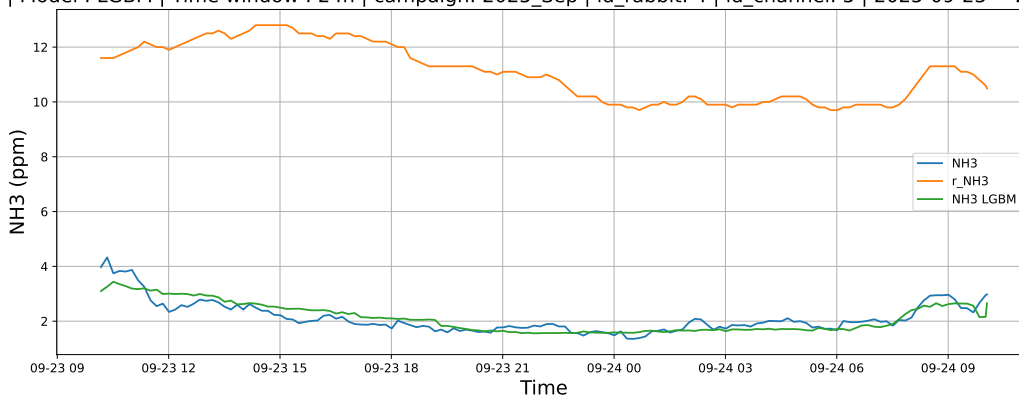
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

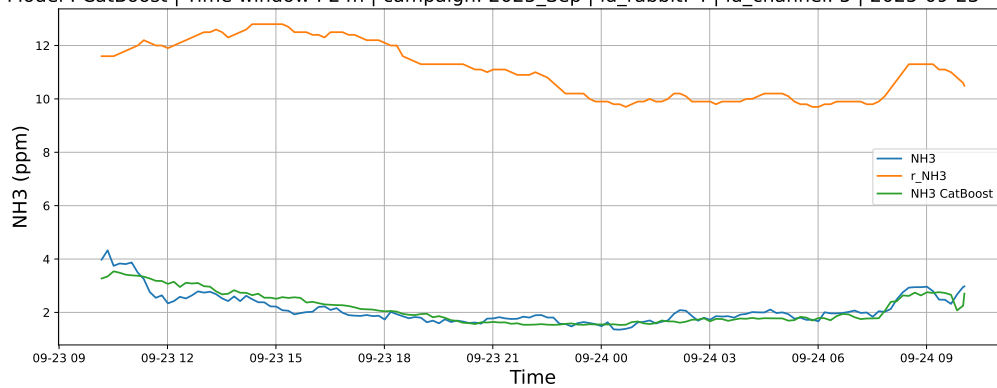
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

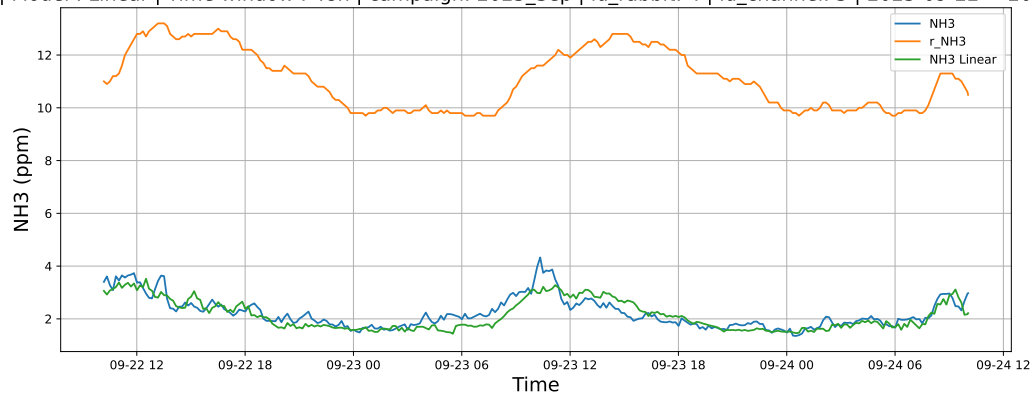


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

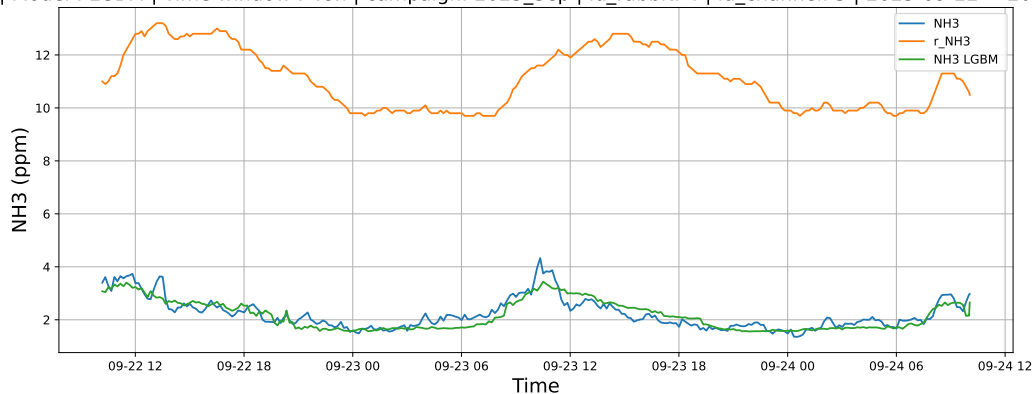


4.1.7.2 Time window : 48

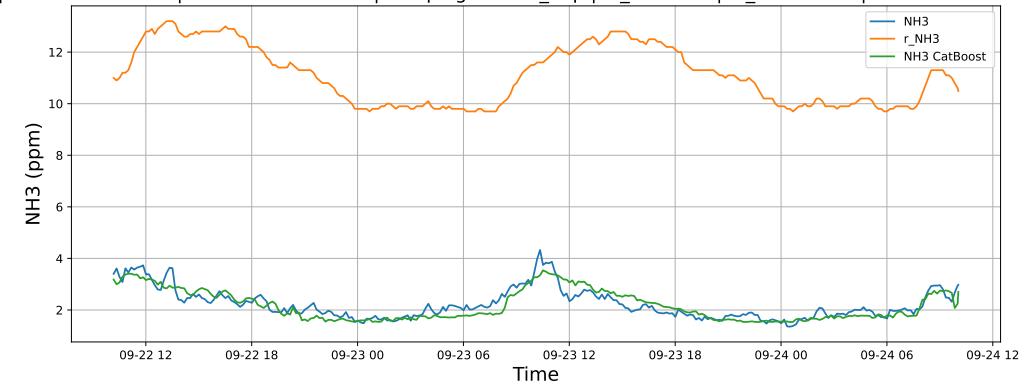
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

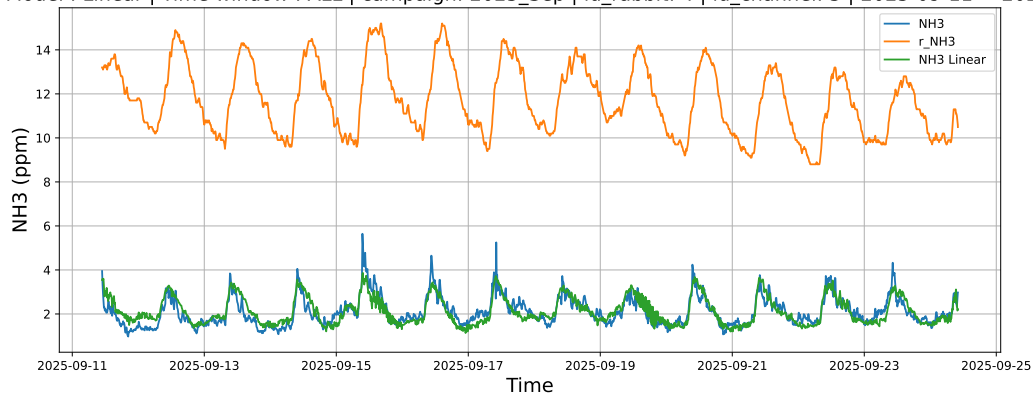


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

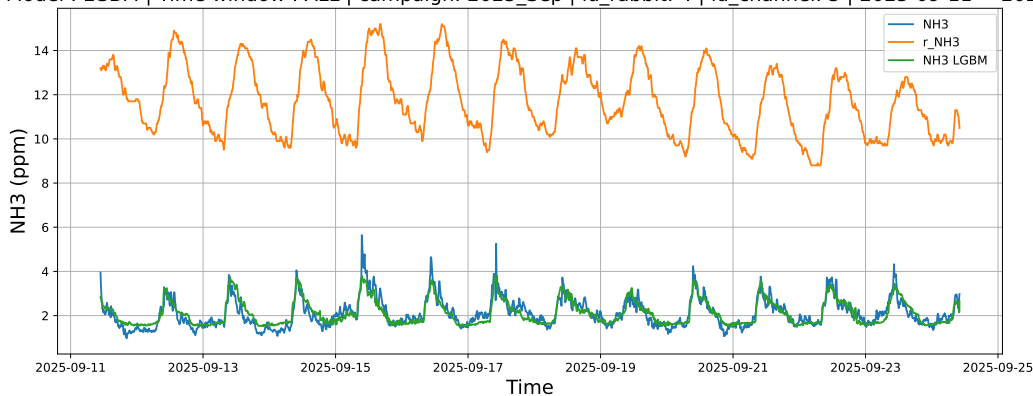


4.1.7.3 Time window : ALL

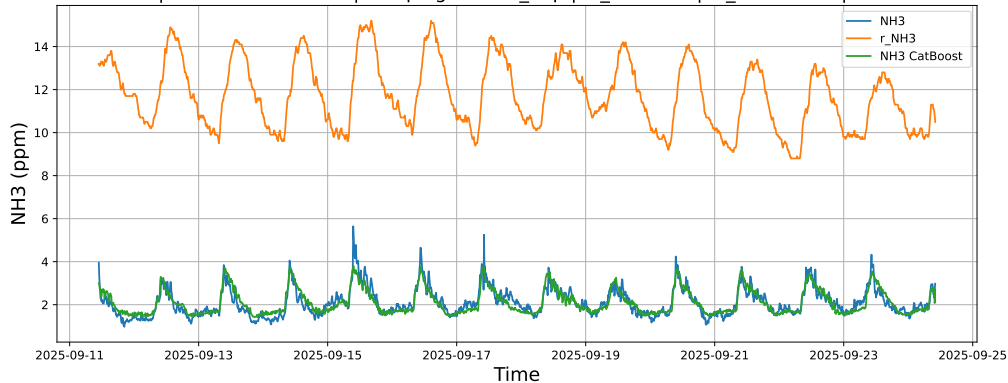
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



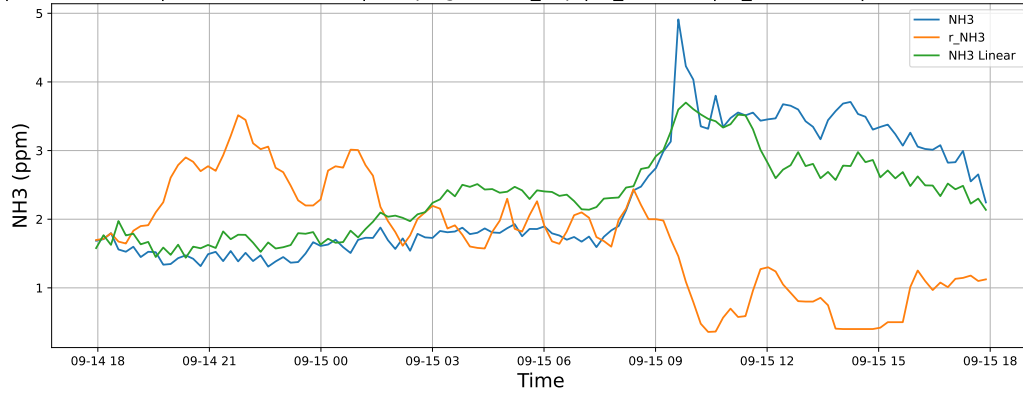
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



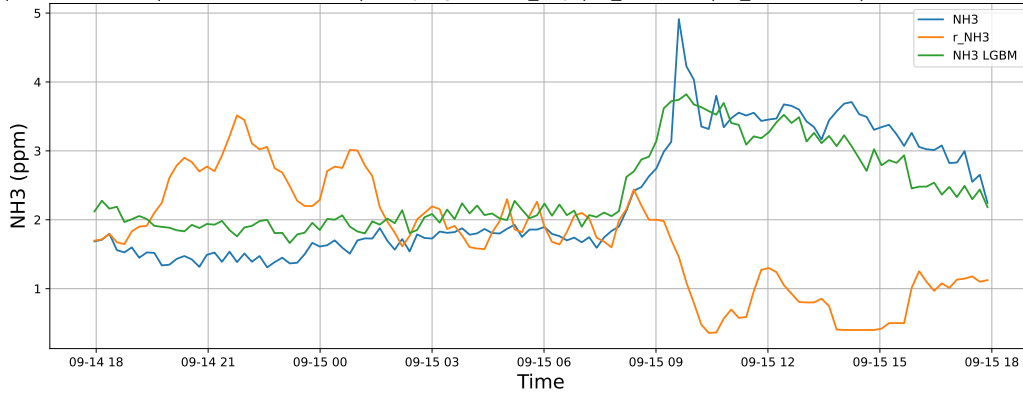
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

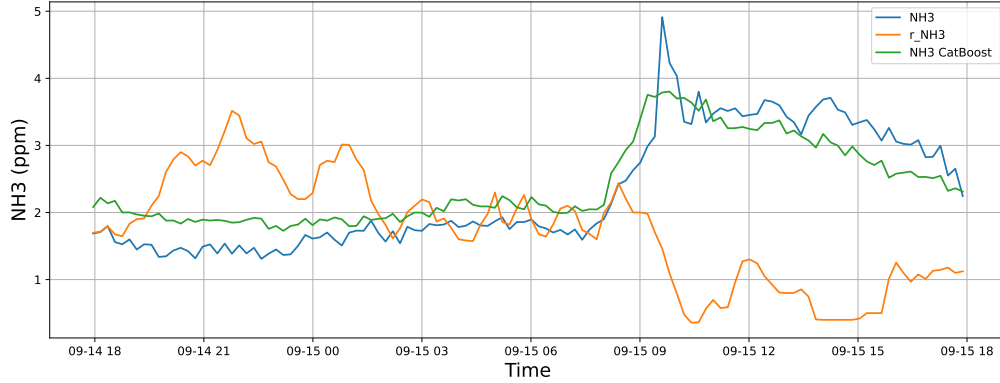
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

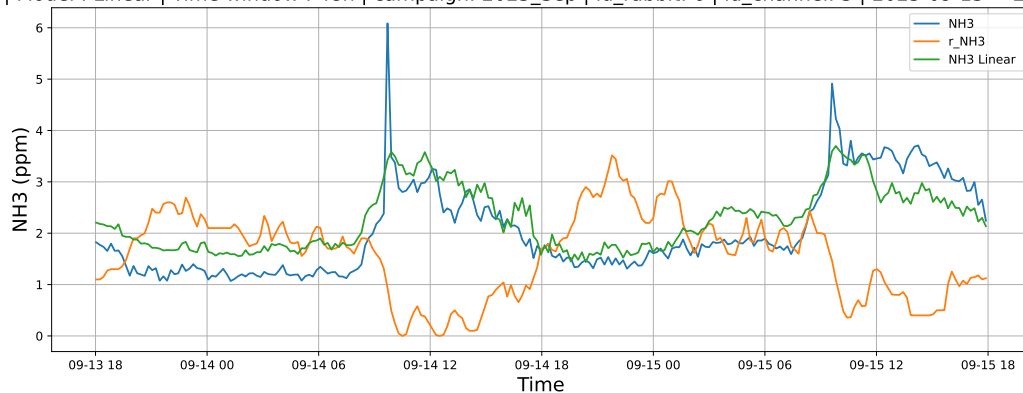


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

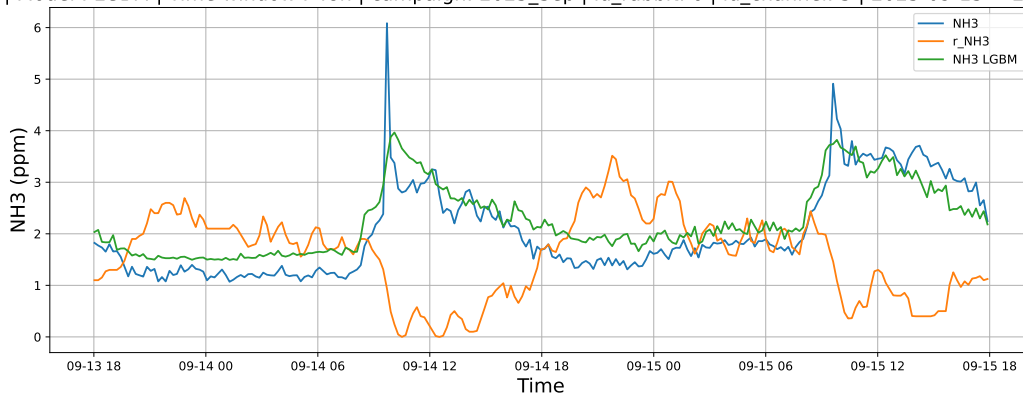


4.1.8.2 Time window : 48

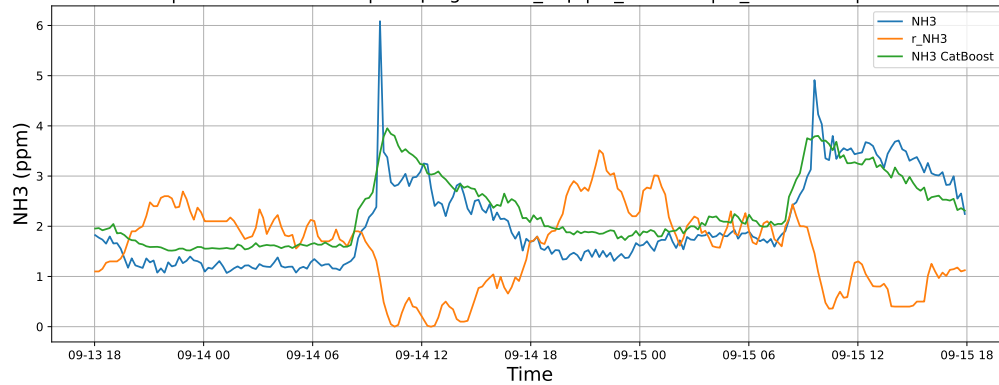
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

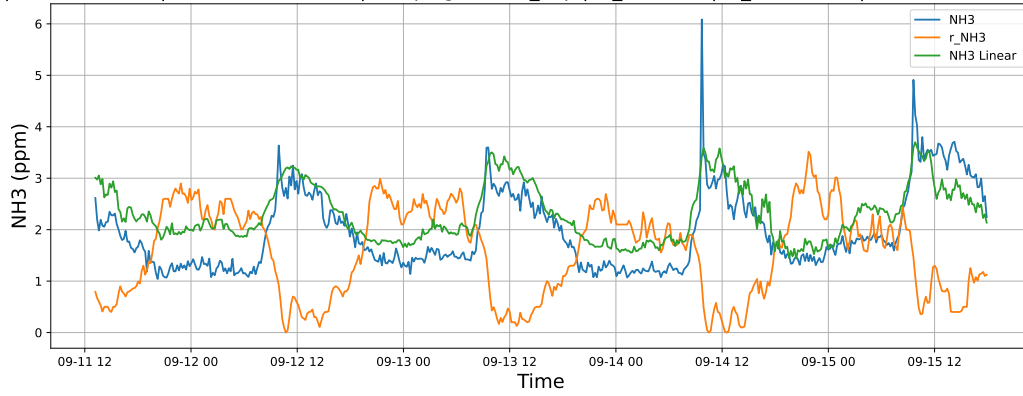


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

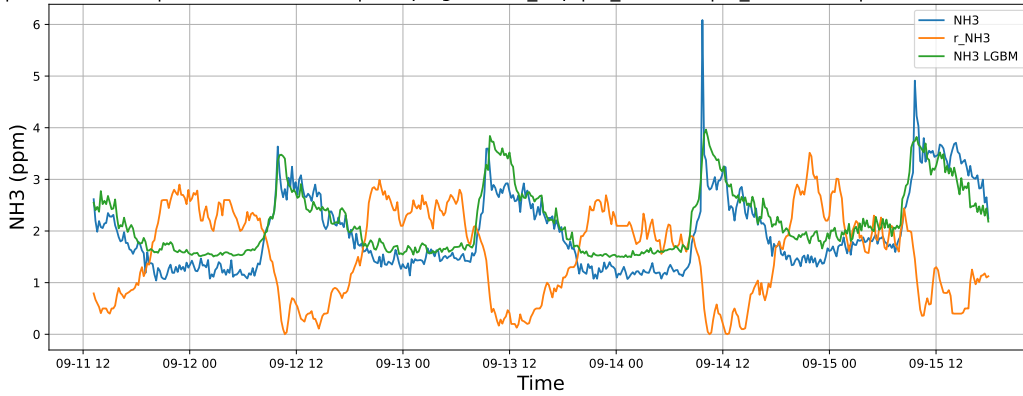


4.1.8.3 Time window : ALL

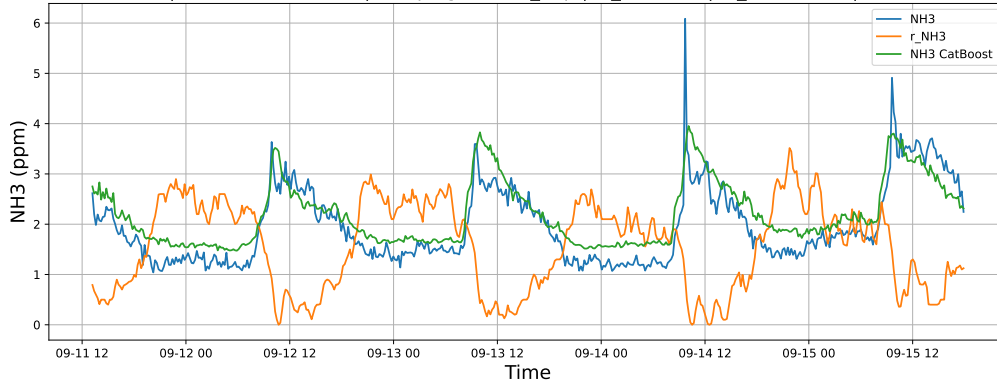
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

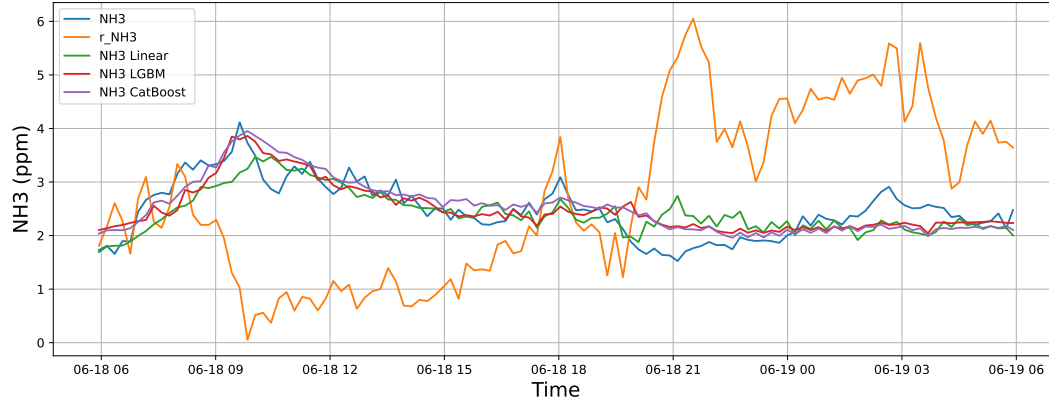


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

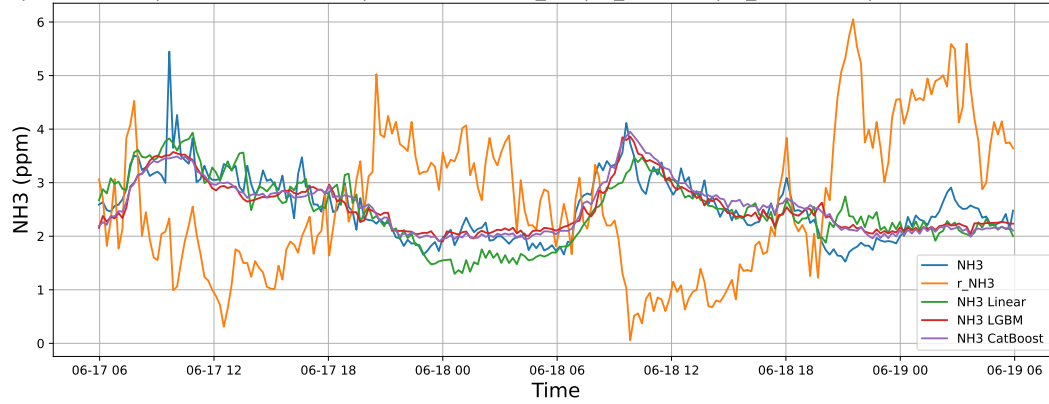
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



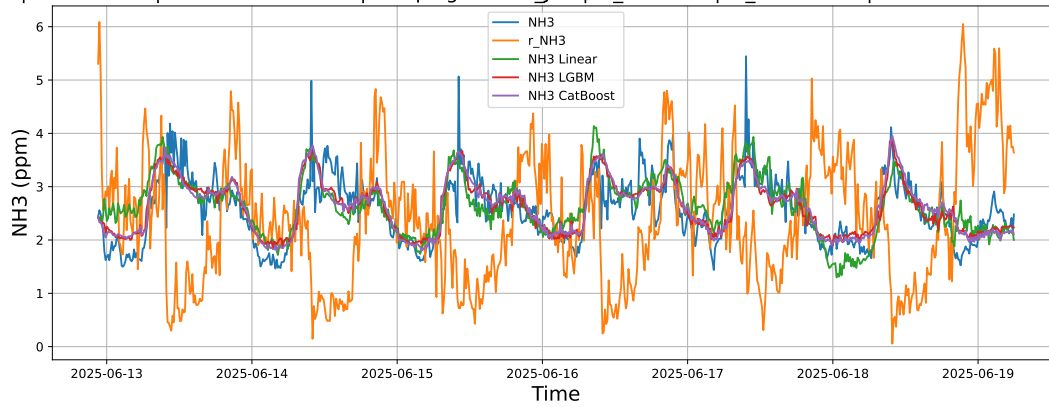
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

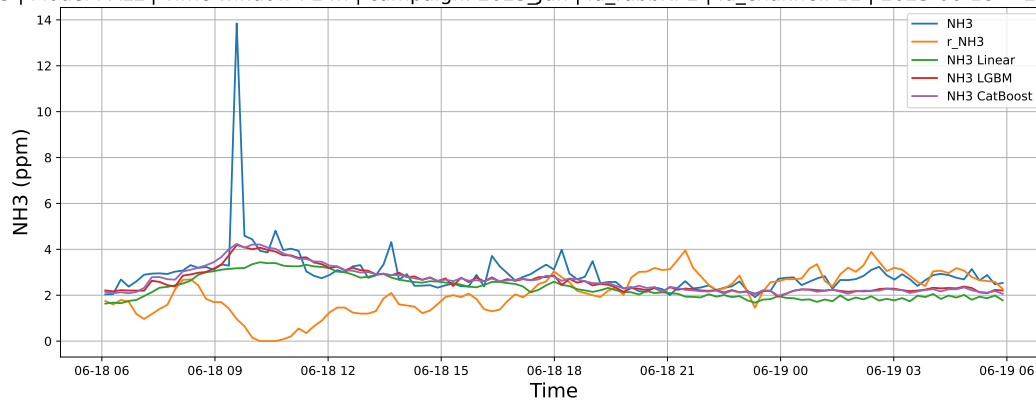
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

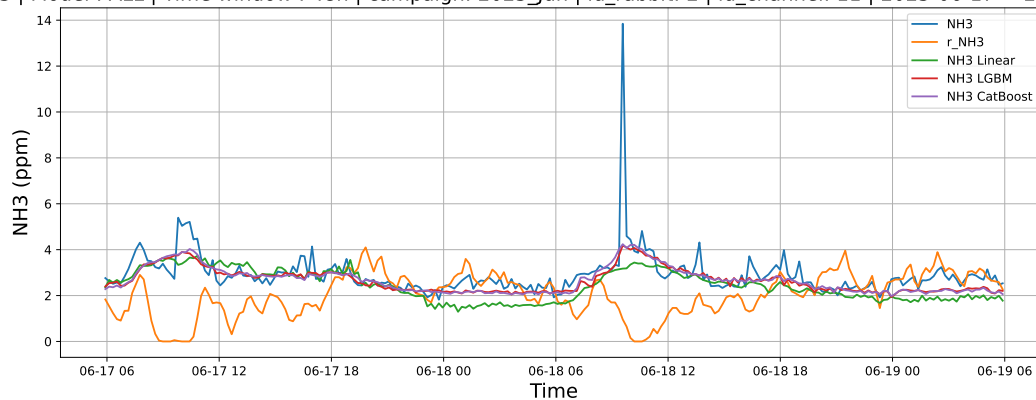
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



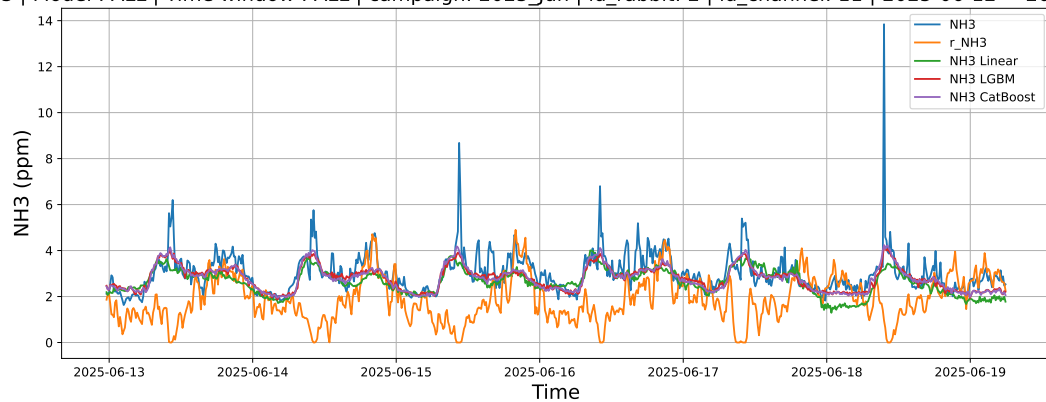
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

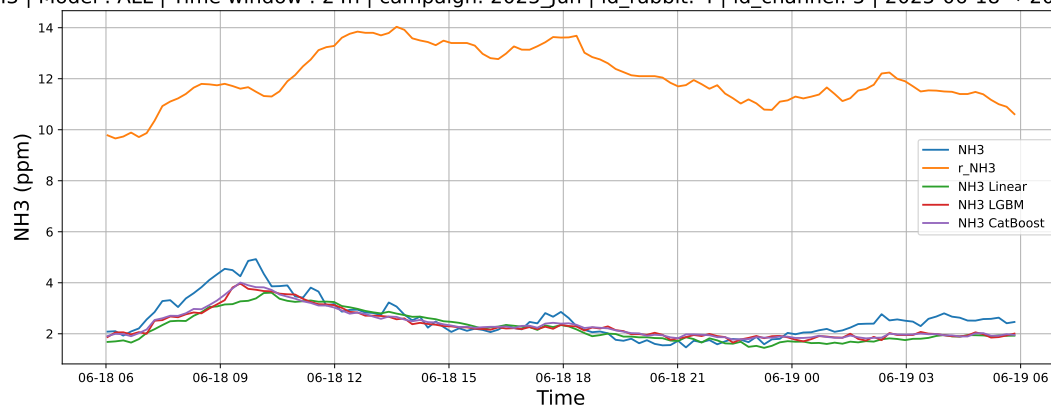
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

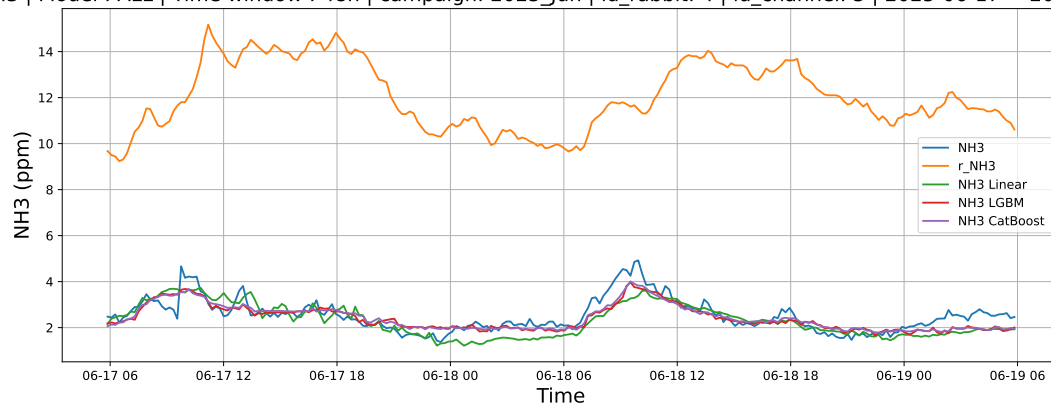
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



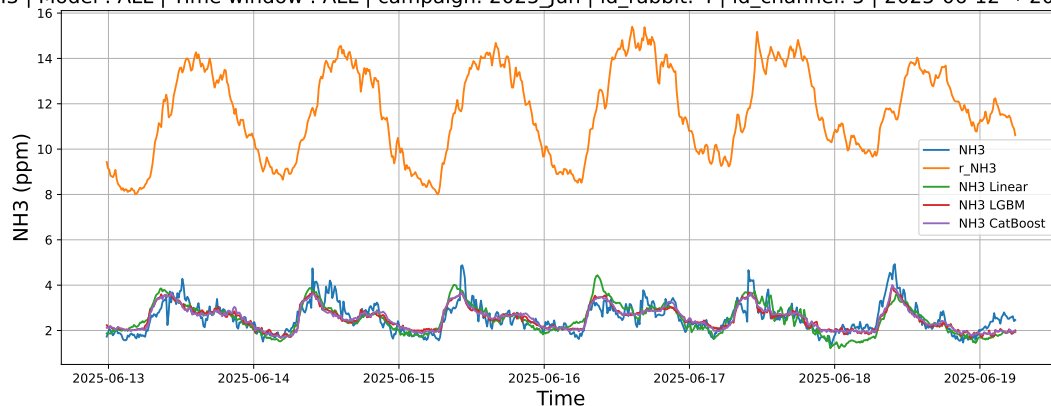
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

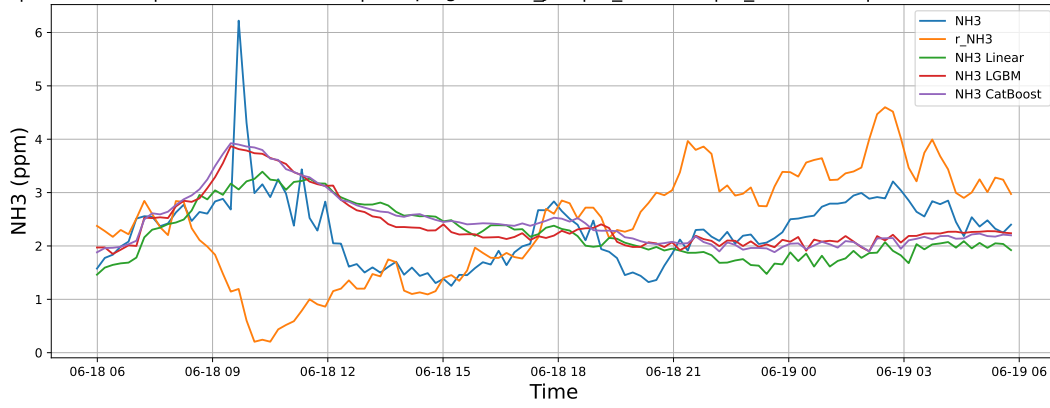
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

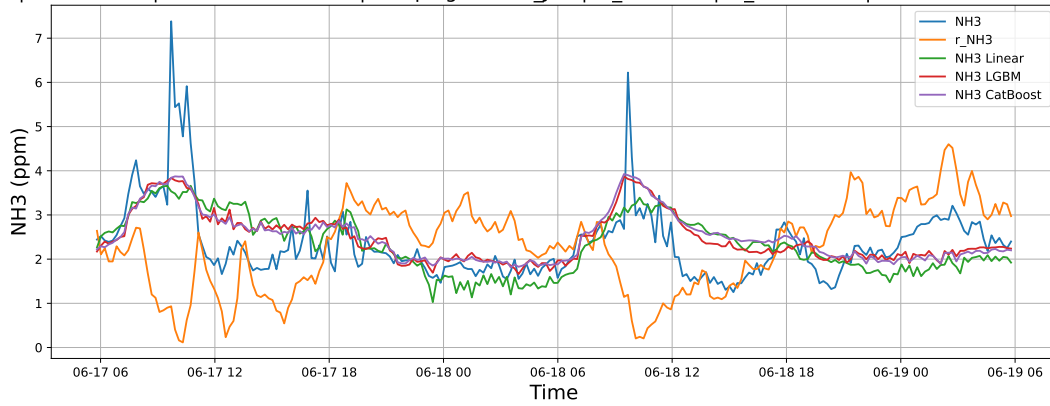
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



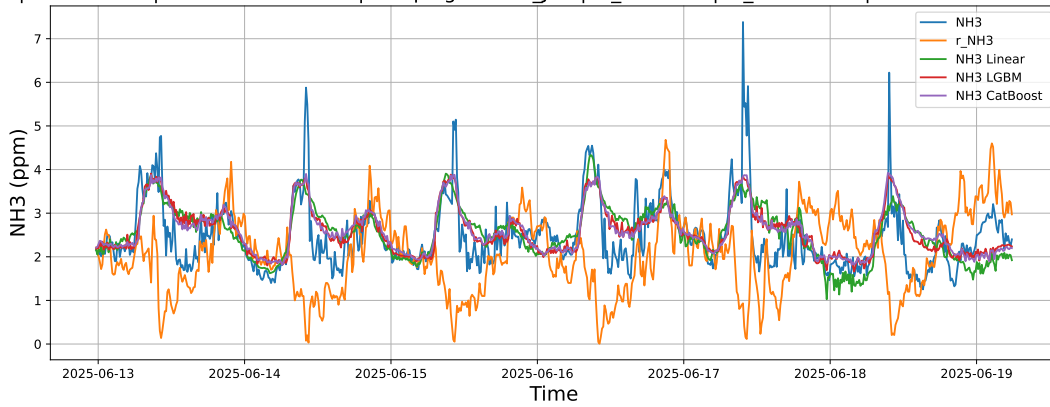
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

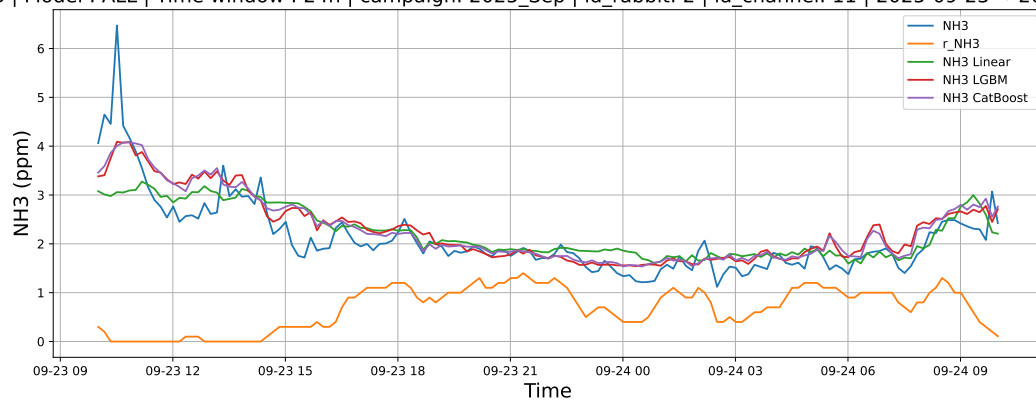
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

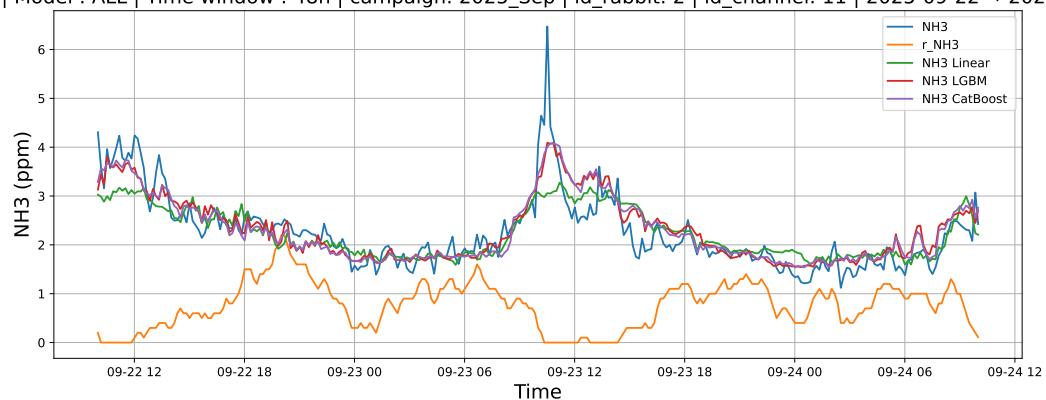
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



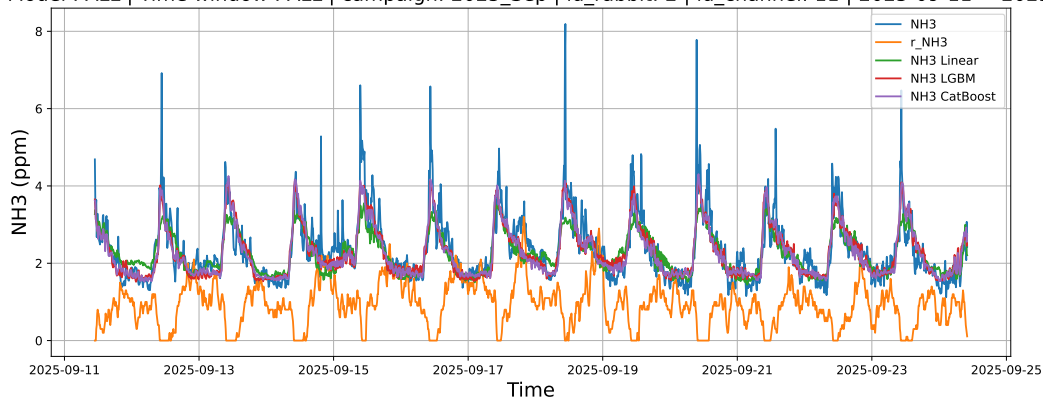
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

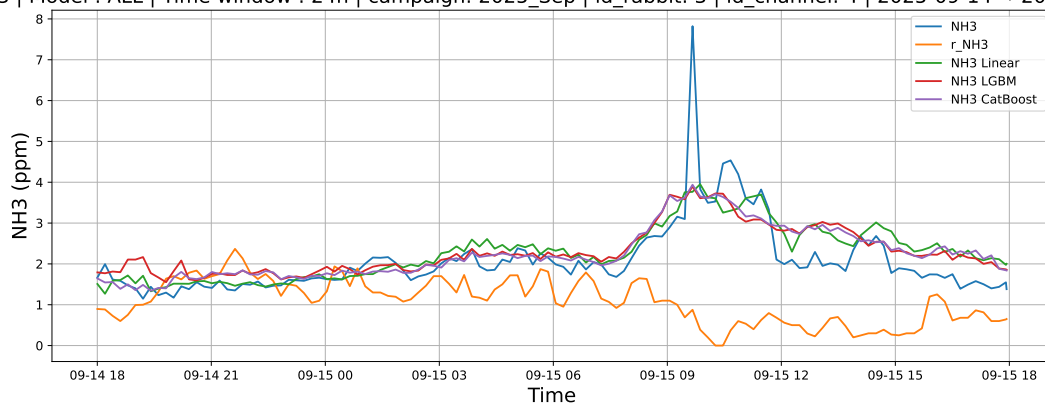
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

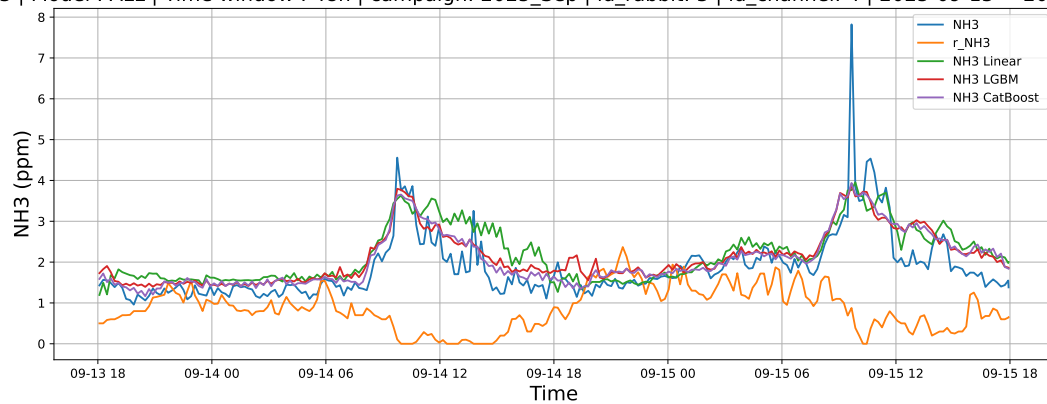
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



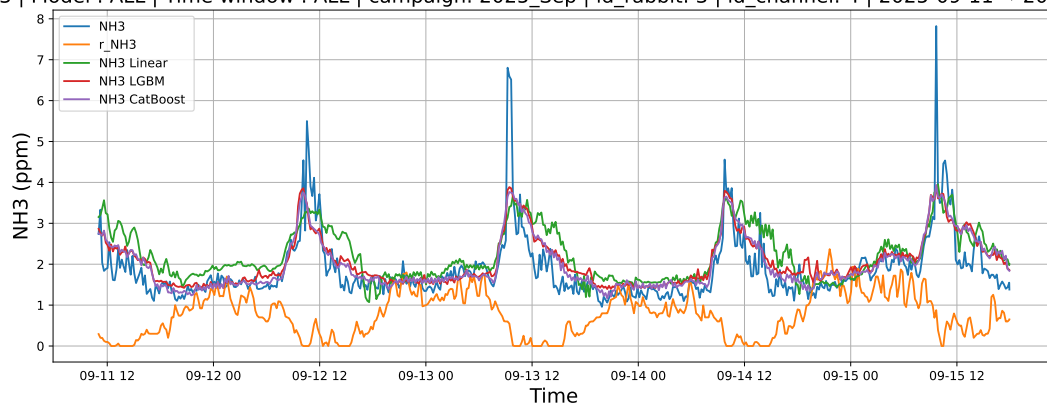
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

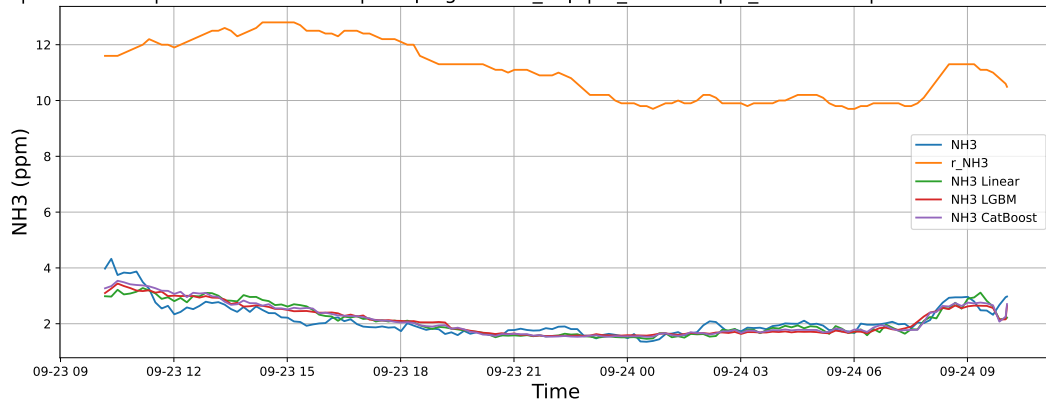
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

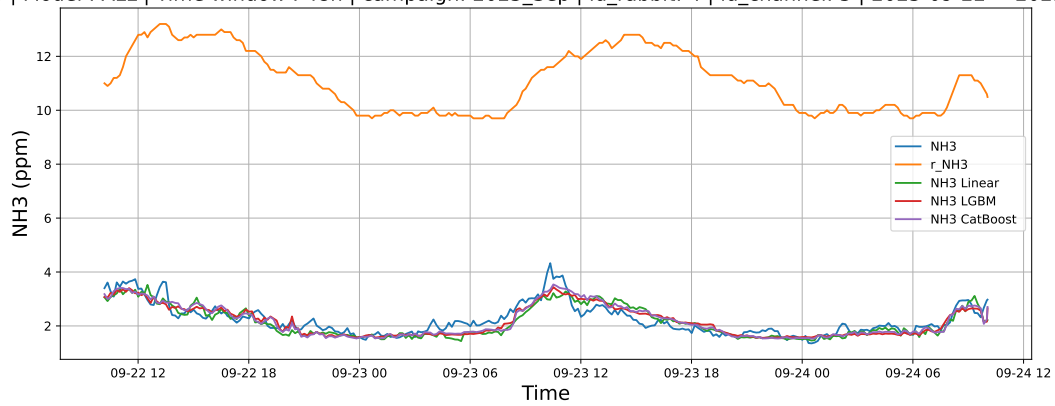
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



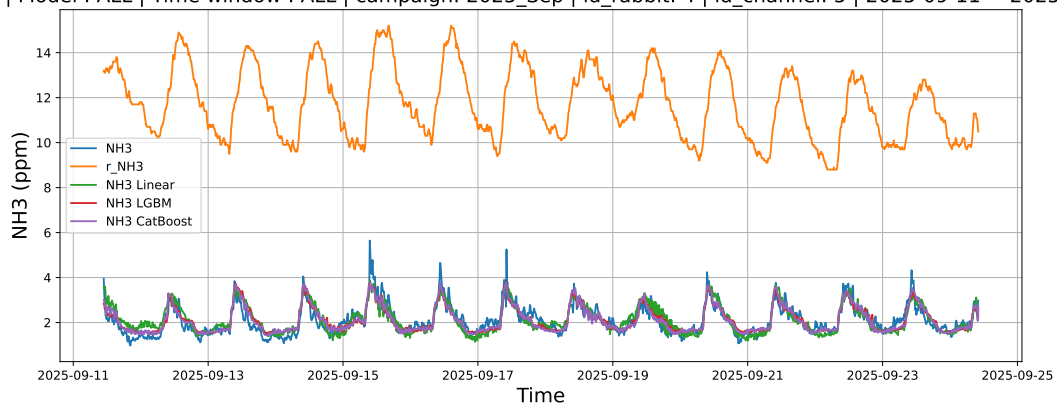
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

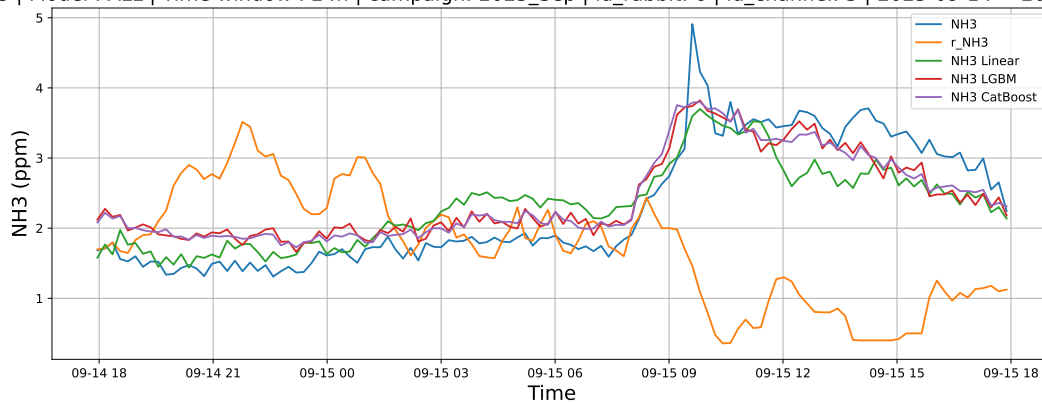
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

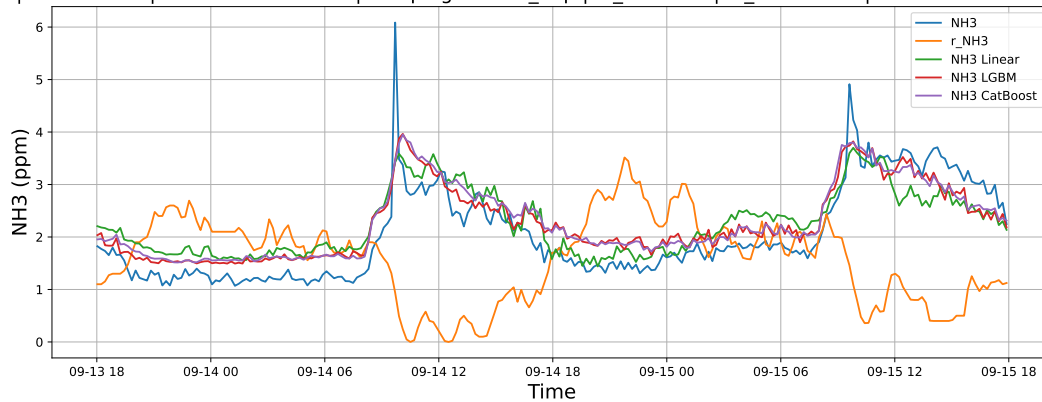
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

